



ELECTRICITY SUPPLY INDUSTRY

STANDARD 43—10

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11 kV SINGLE CIRCUIT OVERHEAD LINES OF LIGHT CONSTRUCTION ON WOOD POLES

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11 kV SINGLE CIRCUIT OVERHEAD LINES OF LIGHT CONSTRUCTION ON WOOD POLES

FOREWORD

Since 1946 the Electricity Supply Industry has made extensive use of BS 1320, 'High Voltage Overhead Lines on Wood Poles for Line Voltages up to and including 11 kV'. The advent of metrication and the coming into force of The Electricity (Overhead Lines) Regulations, 1970, created the need for a revision of BS 1320. Accordingly, this Standard has been prepared to fulfil the needs of the Electricity Supply Industry. It reflects modern practice within the ESI by omitting single phase construction and wooden crossarms and by the introduction of design features to assist the users of Live Line techniques.

The requirements of users outside the Electricity Supply Industry have been met by the publication of a revised BS 1320.

1. SCOPE

This Standard covers 3 phase 11 kV overhead lines of light construction on wood poles, employing the following conductors:

Hard-drawn Copper	—	16 mm ² and 32 mm ²
Aluminium Alloy and ACSR	—	25 mm ² and 50 mm ²

Where a single phase line is required, the standard of construction should be appropriate to 3 phase operation, the centre conductor being omitted.

Lines to this Standard are designed to the conditions specified in 'The Electricity (Overhead Lines) Regulations 1970'. Where these conditions may be exceeded due to geographical or weather conditions, a modified form of construction, as specified in Appendix A, may be adopted.

2. DEFINITIONS

For the purposes of this Standard the following definitions apply:

Intermediate Support — A support in a straight run of a line on which the conductors are supported on pin insulators.

Section Support — A support in a straight run of a line on which the conductors are made off on either side of the crossarm on tension insulator sets.

Pin Angle Support — A support at which a line deviates and the conductors are supported on pin insulators.

Section Angle Support — A support at which a line deviates and the conductors are made off on either side of the crossarm on tension insulator sets.

Span — The horizontal distance between adjacent supports.

Wind Loading Span — The wind loading span associated with any support is half the sum of the spans adjacent to the support.

Average Span — The arithmetic average of a number of spans.

Basic Span — The span length adopted for sag/tension calculations.

Recommended Span — The average span length in any section to which the line should be planned.

Maximum Span — The maximum permitted length of any span.

Sag — The vertical distance, under any system of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured mid-span.

Auxiliary Equipment — Equipment other than that forming part of this line design which may be placed on supports, such as transformers, fusegear etc.

3. REFERENCES

This Standard makes reference to the following documents:

The Electricity (Overhead Lines) Regulations, 1970, SI 1355.

BS 125 'Hard-drawn Copper and Copper-Cadmium Conductors for Overhead Power Transmission Purposes'.

BS 137, 'Insulators of Ceramic Material or Glass for Overhead Lines with a Nominal Voltage greater than 1000 V'.

Part 1, 'Tests'.

Part 2, 'Requirements'.

BS 183, 'General Purpose Galvanized Steel Wire Strand'.

BS 215, 'Aluminium Conductors and Aluminium Conductors, Steel-reinforced, for Overhead Power Transmission'.

Part 1, 'Aluminium Stranded Conductors'.

BS 729, 'Hot Dip Galvanized Coatings on Iron and Steel Articles'.

BS 913, 'Wood Preservation by means of Pressure Creosoting'.

BS 1990, 'Wood Poles for Overhead Lines (Power and Telecommunication Lines)'.

Part 2, 'Metric Units'.

BS 3242, 'Aluminium Alloy Stranded Conductors for Overhead Power Transmission'.

BS 3288, 'Insulator and Conductor Fittings for Overhead Power Lines'.

Part 1, 'Performance and General Requirements'.

Part 2, 'Dimensions'.

BS 4190, 'ISO Metric Black Hexagon Bolts, Screws and Nuts'.

BS 6485, 'PVC-covered Conductors for Overhead Power Lines'.

*ESI Standard 43–90, 'Anti-climbing Devices'.

*ESI Standard 43–91, 'Stay Fittings for Overhead Lines'.

*ESI Standard 43–92, 'Conductor Terminations, Joints and Insulator Binds'.

*ESI Standard 43–93, 'Line Insulators'.

- *ESI Standard 43-95, 'Steelwork'.
- *ESI Standard 43-96, 'Fasteners and Washers'.
- *ESI Standard 43-97, 'Notices and Marker Posts'.
- *ESI Standard 43-100, 'Auxiliary Equipment Erected on Overhead Line Supports'.

* In course of preparation

ESI Standards are revised when necessary by the issue either of revised pages or complete new editions. It is important that users of ESI Standards should ascertain that they are in possession of the latest issues.

4. DESIGN DATA

This Standard is designed in accordance with the requirements of The Electricity (Overhead Lines) Regulations, 1970.

4.1 Conductors

The factor of safety on conductors is 2.5 for all conductors on their nominal breaking load. The conductor tensions at 5°C in still air do not exceed the accepted vibration limits of 20 per cent for aluminium based conductors or 33 per cent for copper conductors of the appropriate breaking loads. In the preparation of sag/tension tables a single basic span has been adopted for each conductor size. The conventional method of deriving a basic span for a section of line has not been adopted because this is considered too precise for practical purposes on wood pole lines.

4.2 Crossarms

The minimum factor of safety on all crossarms is 2.5 on the ultimate tensile strength of the steel. All crossarms are designed to support conductors with a downpull each side of the support corresponding to a gradient of 1 in 10 measured between adjacent pole tops on the recommended span length. Due to the variation between the weights of the conductors employed in this Standard and other considerations, it may be possible to exceed this gradient in certain cases. In these cases calculations should be made. To this end, Table 23 shows the maximum weight which may be accommodated at the ends of each type of crossarm.

4.3 Supports

Intermediate and section support sizes have been related to the wind loadings imposed by the conductors, the loading points being considered to be 150 mm above pole tops. Stayed supports have been designed in accordance with crippling loads derived from conductor loadings. The strength of supports is derived from information contained in BS 1990 with respect to red fir poles. The factor of safety for supports in normal conditions is 2.5.

5. MATERIALS

All components used in the construction of lines to this Standard shall comply with British Standards and/or ESI Standards indicated.

5.1 Conductors

Hard-drawn copper conductors	—	BS 125
Aluminium alloy conductors	—	BS 3242
ACSR	—	BS 215, Part 2
PVC-covered conductors (Type 16)	—	BS 6485

5.2 Wood Poles

Poles shall be best grade European redwood with a minimum ultimate extreme fibre stress of 53.8 N/mm^2 , complying with the requirements of BS 1990, Part 2. They shall be fabricated to Fig. 1, all holes being drilled to template, the centre line of the template being placed along the effective centre line of the pole. Pole markings shall be made on the same side of the pole as that on which the drilling templates have been used. Pole tops shall be evenly quartered, the angle at the apex being 120° . When fully seasoned and fabricated, poles shall be creosoted in accordance with BS 913 by the Rueping Process.

5.3 Crossarm Steelwork

Crossarm steelwork shall comply with ESI Standard 43–95.

5.4 Insulators

String insulator units and pin insulators shall comply with BS 137, Parts 1 and 2, and with ESI Standard 43–93.

5.5 Insulator Fittings and Pins

Insulator fittings and pins shall comply with BS 3288, Parts 1 and 2.

5.6 Conductor Terminations, Joints and Binders

Conductor terminations, joints and binders shall comply with ESI Standard 43–92.

5.7 Stay Strand

Stay strand shall be galvanized steel strand to BS 183. Standard sizes shall be:

Grade 700	—	7/4.0 mm, 19/3.55 mm
Grade 1150	—	7/3.25 mm, 7/4.0 mm

5.8 Stay Components

The following stay components shall comply with ESI Standard 43-91:

- Stay make-offs
- Stay rods
- Stay insulators
- Stay blocks
- Stay thimbles
- Stay shackles

5.9 Bolts

Bolts shall be ISO Metric black hexagon headed to BS 4190, strength grade 4.6. Bolts, nuts and washers shall be galvanized to meet the requirements of BS 729, including brush galvanizing of the screwed portion, with the exception of internal screw threads which shall be oiled. Design calculations are based on a safe working load of 118 N/mm^2 . Flat round and square curved washers and eyebolts shall comply with ESI Standard 43-96.

6. TECHNICAL REQUIREMENTS FOR THE ERECTION OF LINES

6.1 General

Supports shall be erected in accordance with Fig. 2 to 8.

6.2 Maximum and Recommended Span Lengths

Maximum and recommended span lengths are specified in Table 1.

6.3 Wind Loading Spans

Wind loading spans for intermediate and section poles are given in Table 2 (bare conductors) and Table 3 (PVC-covered conductors).

6.4 Angles of Line Deviation

The maximum angles of line deviation on pin insulators are given in Table 5 (bare conductors) and Table 6 (PVC-covered conductors). The maximum angle of line deviation on tension insulator sets is 75° .

6.5 Staying Arrangements and Strut Loading for Angle and Terminal Supports

The components for stays and the application of gusset plates are indicated in the notes on Tables 7 and 8. Staying arrangements are given in Table 7 (bare conductors) and Table 8 (PVC-covered conductors). Stay assemblies shall be to ESI Standard 43-91. When a change of conductor type or size creates out of balance tension on a section pole balancing stays should be fitted on both sides of the pole. The strut loading imposed upon angle and terminal supports shall be checked from Charts 1-12 as appropriate. These loadings shall then be related to the safe working loads of the poles by reference to Table 4.

6.6 Conductor Design Sags and Tensions

Conductor design sags and tensions are given in Tables 11–22. Ordinates for plotting uplift curves and ground clearance curves are given in Table 9 (bare conductors) and Table 10 (PVC-covered conductors). It is necessary to compensate for the initial stretch of new conductor by pre-stressing or overtensioning.

6.7 Use of PVC-covered Conductors

Technical data is provided in this Standard permitting the erection of full section(s) of line conductored with PVC-covered conductors. When a single span of PVC-covered conductor is erected in a section of line, the span shall not exceed 40 per cent (copper conductors) or 60 per cent (aluminium based conductors) of the normal maximum span for the particular bare conductor.

6.8 Provision of Danger Plates

Each support on a line to this Standard shall be provided with a danger plate to ESI Standard 43–97. The plate shall be mounted 3 m above ground level.

6.9 Precautions against Access

The types of support requiring precautions against access and the methods to be employed are detailed in ESI Standard 43–90.

6.10 Clearances

All conductors shall be erected and maintained at the minimum heights specified in The Electricity (Overhead Lines) Regulations, 1970, Clause 7. Data is given in this Standard for conductors operating at a maximum temperature of 50°C (Tables 9 and 10).

ACSR and aluminium alloy conductors shall be erected with an additional clearance of 450 mm above those specified in columns 2 and 3 of Schedule 1 of the Regulations.

6.11 Erection of Auxiliary Equipment

Where a support carries auxiliary equipment, the construction of the support shall be in accordance with ESI Standard 43–100 which also contains details of the earthing of these supports.

No allowance has been made in the basic support considerations for the additional mechanical loadings which would be imposed by auxiliary equipment and additional conductors.

6.12 Relationships with other Bodies

Where lines to this Standard are erected over or alongside the plant of the following Bodies the provisions indicated shall be complied with:

The Post Office	—	Post Office Memorandum A231
British Rail	—	The Second Schedule (General and Engineering Conditions) of the Railway Master Wayleave Agreement

Additionally, the requirements of the following Body shall be noted:

British Waterways Board	—	The Schedule of Minimum Clearances above Waterways and Reservoirs
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Table 1 -- Recommended and Maximum Span Lengths

Conductors	Recommended Span (m)	Maximum Span (m)
16 mm ² Copper 25 mm ² Aluminium Alloy and ACSR	90	110
32 mm ² Copper 50 mm ² Aluminium Alloy and ACSR	125	140

Table 2 -- Wind Loading Spans for Intermediate and Section Poles -- Bare Conductors

Pole Details			Wind Loading Spans (m)					
Length (m)	Planting Depth (m)	Dia. at 1.5 m from butt (mm)	3 x 16 mm ² Copper	3 x 32 mm ² Copper	3 x 25 mm ² Aluminium Alloy	3 x 50 mm ² Aluminium Alloy	3 x 25 mm ² ACSR	3 x 50 mm ² ACSR
8.5	1.5	180 (L)	110	94	108	76	107	75
		190	—	110	110	90	110	88
		200	—	127	—	103	—	102
		210	—	140	—	116	—	115
		215 (M)	—	—	—	130	—	128
		225	—	—	—	140	—	140
9.0	1.5	180 (L)	110	88	101	71	100	70
		190	—	103	110	84	110	83
		200	—	119	—	96	—	95
		210	—	134	—	109	—	107
		220 (M)	—	140	—	130	—	128
		230	—	—	—	140	—	140
9.5	1.5	185 (L)	110	89	102	73	102	72
		190	—	97	110	79	110	77
		200	—	111	—	90	—	89
		210	—	126	—	102	—	101
		220	—	140	—	114	—	112
		225 (M)	—	—	—	131	—	129
10.0	1.5	235	—	—	—	140	—	140
		185 (L)	110	84	97	68	96	67
		190	—	91	105	74	104	73
		200	—	105	110	85	110	84
		210	—	119	—	96	—	95
		220	—	132	—	107	—	106
10.5	1.5	230 (M)	—	140	—	131	—	129
		240	—	—	—	140	—	140
		190 (L)	110	86	99	70	98	69
		200	—	99	110	81	110	79
		210	—	112	—	91	—	90
		220	—	125	—	102	—	100
10.5	1.5	230	—	138	—	112	—	110
		235 (M)	—	140	—	132	—	130
		245	—	—	—	140	—	140

(Cont'd)

Table 2 (Cont'd)

Pole Details			Wind Loading Spans (m)					
Length (m)	Planting Depth (m)	Dia. at 1.5 m from butt (mm)	3 x 16 mm ² Copper	3 x 32 mm ² Copper	3 x 25 mm ² Aluminium Alloy	3 x 50 mm ² Aluminium Alloy	3 x 25 mm ² ACSR	3 x 50 mm ² ACSR
11.0	1.8	195 (L)	110	88	101	71	100	70
		200	—	94	108	76	107	75
		210	—	106	110	86	110	85
		220	—	119	—	96	—	95
		230	—	131	—	106	—	105
		240 (M)	—	140	—	132	—	130
		250	—	—	—	140	—	140
11.5	1.8	245 (M)	110	140	110	132	110	131
		255	—	—	—	140	—	140
12.0	1.8	250 (M)	110	140	110	133	110	131
		260	—	—	—	140	—	140
13.0	1.8	260 (M)	110	140	110	138	110	136
		270	—	—	—	140	—	140
14.0	1.8	275 (M)	110	140	110	140	110	140
15.0	2.1	290 (M)	110	140	110	140	110	140
16.0	2.1	305 (M)	110	140	110	140	110	140

Notes:

1. (L) or (M) against a pole diameter indicates that the pole is the standard light or medium pole in accordance with BS 1990.
2. Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Table 3 – Wind Loading Spans for Intermediate and Section Poles – PVC-covered Conductors

Pole Details			Wind Loading Spans (m)					
Length (m)	Planting Depth (m)	Dia. at 1.5 m from butt (mm)	3 x 16 mm ² Copper	3 x 32 mm ² Copper	3 x 25 mm ² Aluminium Alloy	3 x 50 mm ² Aluminium Alloy	3 x 25 mm ² ACSR	3 x 50 mm ² ACSR
8.5	1.5	180 (L)	81	62	71	54	70	53
		190	95	73	83	63	83	63
		200	110	84	96	73	95	72
		210	—	95	109	83	108	82
		215 (M)	—	106	110	92	110	91
		225	—	122	—	106	—	104
		235	—	138	—	120	—	118
		245	—	140	—	134	—	132
		255	—	—	—	140	—	140
9.0	1.5	180 (L)	75	58	66	50	66	50
		190	89	68	78	59	77	58
		200	103	79	90	69	89	68
		210	110	89	102	78	101	76
		220 (M)	—	106	110	92	110	91
		230	—	121	—	106	—	104
		240	—	136	—	119	—	117
		250	—	140	—	131	—	130
		260	—	—	—	140	—	140
9.5	1.5	185 (L)	77	59	67	51	67	51
		190	83	64	73	56	72	55
		200	96	74	84	64	84	64
		210	109	84	95	73	95	72
		220	110	93	106	81	105	80
		225 (M)	—	107	110	93	110	91
		235	—	121	—	105	—	104
		245	—	135	—	117	—	116
		255	—	140	—	129	—	128
10.0	1.5	265	—	—	—	140	—	139
		275	—	—	—	—	—	140
		185 (L)	72	56	64	48	63	48
		190	79	60	69	52	68	52
		200	91	70	80	61	79	60
		210	103	79	90	69	89	68
		220	110	88	100	76	99	75
		230 (M)	—	107	110	93	110	92
		240	—	121	—	105	—	103
10.5	1.5	250	—	134	—	116	—	115
		260	—	140	—	128	—	126
		270	—	—	—	139	—	137
		280	—	—	—	140	—	140
		190 (L)	74	57	65	50	64	49
		200	86	66	75	57	75	57
		210	97	75	85	65	84	64
		220	108	83	95	72	94	71
		230	110	92	105	80	104	79
	1.5	235 (M)	—	108	110	94	110	92
		245	—	120	—	105	—	103
		255	—	133	—	115	—	114
		265	—	140	—	126	—	124
		275	—	—	—	137	—	135
		285	—	—	—	140	—	140

(Cont'd)

Table 3 (Cont'd)

Pole Details			Wind Loading Spans (m)					
Length (m)	Planting Depth (m)	Dia. 1.5 m from butt (mm)	3 x 16 mm ² Copper	3 x 32 mm ² Copper	3 x 25 mm ² Aluminium Alloy	3 x 50 mm ² Aluminium Alloy	3 x 25 mm ² ACSR	3 x 50 mm ² ACSR
11.0	1.8	195 (L)	76	58	67	51	66	50
		200	81	63	71	54	71	54
		210	92	71	81	62	80	61
		220	103	79	90	69	89	68
		230	110	87	99	76	98	75
		240 (M)	—	108	110	94	110	93
		250	—	120	—	104	—	103
		260	—	132	—	115	—	113
		270	—	140	—	125	—	123
		280	—	—	—	135	—	133
		290	—	—	—	140	—	140
11.5	1.8	245 (M)	110	108	110	94	110	93
		255	—	120	—	104	—	103
		265	—	131	—	114	—	112
		275	—	140	—	123	—	122
		285	—	—	—	133	—	131
		295	—	—	—	140	—	140
12.0	1.8	250 (M)	110	109	110	95	110	93
		260	—	119	—	104	—	102
		270	—	130	—	113	—	111
		280	—	140	—	122	—	121
		290	—	—	—	131	—	130
		300	—	—	—	140	—	140
13.0	1.8	260 (M)	110	113	110	99	110	97
		270	—	124	—	108	—	107
		280	—	135	—	118	—	116
		290	—	140	—	127	—	126
		300	—	—	—	137	—	135
		310	—	—	—	140	—	140
14.0	1.8	275 (M)	110	120	110	104	110	103
		285	—	130	—	113	—	111
		295	—	140	—	122	—	120
		305	—	—	—	131	—	129
		315	—	—	—	140	—	138
		325	—	—	—	—	—	140
15.0	2.1	290 (M)	110	128	110	112	110	110
		300	—	138	—	120	—	119
		310	—	140	—	129	—	127
		320	—	—	—	138	—	136
		330	—	—	—	140	—	140
16.0	2.1	305 (M)	110	137	110	119	110	117
		315	—	140	—	128	—	126
		325	—	—	—	136	—	134
		335	—	—	—	140	—	140

Notes:

1. (L) or (M) against a pole diameter indicates that the pole is the standard light or medium pole in accordance with BS 1990.
2. Poles are assumed to have the minimum top diameter in accordance with BS 1990.

Table 4 - Permissible Strut Loadings of Poles

Pole Planting Depths (m)	Pole Lengths (m)	Pole Top Dia. (mm)	Max. Strut Loading (kgf)		Pole Top Dia. (mm)	Max. Strut Loading (kgf)		Pole Top Dia. (mm)	Max. Strut Loading (kgf)	Dia. at 1.5 m (mm)	Pole Class (BS 1990)	
1.5	8.5	125	1542		140	1978		150	2315	180	L	
	9.0	125	1337		140	1718		150	2013	180		
	9.5	125	1232		140	1579		160	2150	185		
	10.0	125	1087		140	1395		160	1902	185		
	10.5	125	1016		140	1301		160	1770	190		
1.8	11.0	125	998		140	1282		160	1750	195		
	11.5	125	939		145	1303		165	1763	200		
	12.0	125	847		145	1176		165	1593	200		
	13.0	130	834		150	1144		170	1533	210		
1.5	8.5	150	3171		165	3907		180	4765	215	M	
	9.0	150	2870		165	3533		180	4304	220		
	9.5	150	2620		165	3221		180	3921	225		
	10.0	150	2409		165	2959		185	3833	230		
	10.5	150	2231		170	2923		190	3764	235		
1.8	11.0	150	2169		170	2853		190	3688	240		
	11.5	150	2024		170	2657		190	3428	245		
	12.0	150	1898		170	2487		190	3203	250		
	13.0	160	1936		180	2505		200	3191	260		
2.1	14.0	160	1809		180	2327		205	3123	275		
	15.0	165	1872		185	2394		210	3193	290		
	16.0	170	1887		190	2392		215	3159	305		
1.5	8.5	190	7762		215	10228		240	13239	265	S	
	9.0	190	7209		215	9470		240	12226	275		
	9.5	190	6525		215	8565		245	11604	280		
	10.0	190	5952		215	7805		245	10564	285		
	10.5	190	5466		220	7541		250	10157	290		
1.8	11.0	190	5282		220	7325		250	9910	295		
	11.5	190	4891		220	6771		250	9147	300		
	12.0	190	4553		220	6292		250	8487	305		
	13.0	195	4351		225	5953		255	7961	320		
2.1	14.0	195	3988		225	5426		255	7221	335		
	15.0	195	3801		225	5167		260	7193	350		
	16.0	200	3736		230	5032		265	6939	365		
		Poles with top dia. equal to min. specified in BS 1990.			Poles with mean value of top dia.			Poles with top dia. equal to max. specified in BS 1990.				

Notes:

1. The above values of strut loading are safe working loads.
2. This table shall be used in conjunction with Charts 1-12.

Maximum Angles of Line Deviation of Pin Insulators

Table 5 – Bare Conductors

Type of Pin	Maximum Angle of Line Deviation (°)					
	16 mm ² Copper	32 mm ² Copper	25 mm ² Aluminium Alloy	50 mm ² Aluminium Alloy	25 mm ² ACSR	50 mm ² ACSR
10 kN MFL	45	29	45	20	30	10

Table 6 – PVC-covered Conductors

Type of Pin	Maximum Angle of Line Deviation (°)					
	16 mm ² Copper	32 mm ² Copper	25 mm ² Aluminium Alloy	50 mm ² Aluminium Alloy	25 mm ² ACSR	50 mm ² ACSR
10 kN MFL	45	24	45	17	30	10

Table 7 - Stay Details - Bare Conductors (excluding Pole Strut Loading Considerations)

Conductors	Angle of Stay Slope	Maximum Angles of Line Deviation			
		Grade 700 Stay Strand		Grade 1150 Stay Strand	
		1 Stay 7/4.0 mm	1 Stay 19/3.55 mm or 2 Stays 7/4.0 mm	1 Stay 7/3.25 mm	1 Stay 7/4.0 mm or 2 Stays 7/3.25 mm
3 x 16 mm ² Copper	30°	75°	—	75°	—
	35°	75°	—	75°	—
	40°	75°	—	75°	—
	45°	75°	—	75°	—
3 x 32 mm ² Copper	30°	38°	75°	41°	71°
	35°	45°	75°	48°	75°
	40°	52°	75°	56°	75°
	45°	59°	75°	63°	75°
3 x 25 mm ² Aluminium Alloy	30°	64°	75°	69°	75°
	35°	75°	—	75°	—
	40°	75°	—	75°	—
	45°	75°	—	75°	—
3 x 50 mm ² Aluminium Alloy	30°	27°	56°	29°	51°
	35°	32°	67°	34°	60°
	40°	37°	75°	40°	70°
	45°	42°	75°	45°	75°
3 x 25 mm ² ACSR	30°	56°	75°	60°	75°
	35°	66°	75°	71°	75°
	40°	75°	—	75°	—
	45°	75°	—	75°	—
3 x 50 mm ² ACSR	30°	24°	51°	26°	46°
	35°	29°	61°	31°	54°
	40°	34°	70°	36°	63°
	45°	38°	75°	41°	71°

For notes on Tables 7 and 8 see page 20.

Table 8 – Stay Details – PVC-covered Conductors (excluding Pole Strut Loading Considerations)

Conductors	Angle of Stay Slope	Maximum Angles of Line Deviation			
		Grade 700 Stay Strand		Grade 1150 Stay Strand	
		1 Stay 7/4.0 mm	1 Stay 19/3.55 mm or 2 Stays 7/4.0 mm	1 Stay 7/3.25 mm	1 Stay 7/4.0 mm or 2 Stays 7/3.25 mm
3 x 16 mm ² Copper	30°	75°	—	75°	—
	35°	75°	—	75°	—
	40°	75°	—	75°	—
	45°	75°	—	75°	—
3 x 32 mm ² Copper	30°	33°	74°	36°	66°
	35°	41°	75°	44°	75°
	40°	47°	75°	51°	75°
	45°	54°	75°	58°	75°
3 x 25 mm ² Aluminium Alloy	30°	60°	75°	64°	75°
	35°	72°	75°	75°	—
	40°	75°	—	75°	—
	45°	75°	—	75°	—
3 x 50 mm ² Aluminium Alloy	30°	23°	53°	25°	47°
	35°	29°	63°	31°	56°
	40°	34°	74°	36°	66°
	45°	39°	75°	41°	75°
3 x 25 mm ² ACSR	30°	52°	75°	55°	75°
	35°	62°	75°	66°	75°
	40°	72°	75°	75°	—
	45°	75°	—	75°	—
3 x 50 mm ² ACSR	30°	21°	48°	23°	43°
	35°	26°	57°	28°	51°
	40°	31°	66°	33°	59°
	45°	35°	75°	38°	67°

For notes on Tables 7 and 8 see page 20.

Notes on Tables 7 and 8 – Stay Details

1. **Terminal Stays.** Terminal stays shall be in accordance with the corresponding requirements for 60° angles of deviation.
2. **Strut Loading of Poles.** Poles suitable for combinations of stay slopes and angles of deviation are listed in Table 4.
3. **Gusset Plates** are required when the angles of line deviation exceed the following:

50 mm² ACSR – PVC Insulated — 56° Determined by the loading
on the crossarm/pole bolts.

All other conductors — 60° Determined by crossarm
mounting position.

4. **Stay Components** shall be in accordance with the following table. (Six figure numbers refer to the ESI drawing number.)

Stay Strand	Stay Rod	Stay Insulator	Stay Block	MFL of assembly (kgf)	Components determining MFL
Grade 700 7/4.0 mm	439101 Type 1	439106 Beam Type 1 or 439107 Porcelain Type 2	439103 Type 1 or 439102	6281	Stay strand
Grade 700 19/3.55 mm	439101 Type 2	439106 Beam Type 2 or 439107 Porcelain Type 2	439103 Type 2	11216	Rod Insulator Block
Grade 1150 7/3.25 mm	439101 Type 1	439106 Beam Type 1 or 439107 Porcelain Type 2	439103 Type 1 or 439102	6628	Rod Block
Grade 1150 7/4.0 mm	439101 Type 2	439106 Beam Type 2 or 439107 Porcelain Type 2	439103 Type 2	10299	Stay strand

When a stay strap (Drawing No. 439110) is used the SWL for the stay must be restricted to 2957 kgf.

Table 9 – Ordinates for Plotting Uplift and Ground Clearance Templates – Bare Conductors

1. 16 mm² Copper and 25 mm² Aluminium Alloy and ACSR (Basic Span 90 m)

Span (m)	Sag (m)					
	16 mm ² Cu		25 mm ² Al. Alloy		25 mm ² ACSR	
	-6°C (Still Air) Cold Curve	50°C Hot Curve	-6°C (Still Air) Cold Curve	50°C Hot Curve	-6°C (Still Air) Cold Curve	50°C Hot Curve
25	0.08	0.14	0.03	0.10	0.03	0.10
50	0.34	0.56	0.13	0.41	0.14	0.39
75	0.76	1.26	0.29	0.93	0.31	0.87
100	1.35	2.24	0.52	1.65	0.55	1.54
125	2.10	3.50	0.81	2.58	0.87	2.41
150	3.03	5.03	1.16	3.72	1.25	3.47
175	4.12	6.85	1.58	5.07	1.70	4.73
200	5.39	8.95	2.06	6.62	2.22	6.17
250	8.41	13.99	3.23	10.34	3.46	9.65
300	12.12	20.14	4.64	14.89	4.99	13.89
350	16.49	27.41	6.32	20.26	6.79	18.91
400	21.54	35.80	8.26	26.46	9.86	24.69
450	27.26	45.31	10.45	33.49	11.22	31.25
500	33.66	55.94	12.90	41.35	13.85	38.58
550	40.73	67.69	15.61	50.03	16.76	46.69
600	48.47	80.56	18.58	59.54	19.95	55.56

Table 9 (Cont'd)

2. 32 mm² Copper and 50 mm² Aluminium Alloy and ACSR (Basic Span 125 m)

Span (m)	Sag (m)					
	32 mm ² Cu		50 mm ² Al. Alloy		25 mm ² ACSR	
	–6°C (Still Air) Cold Curve	50°C Hot Curve	–6°C (Still Air) Cold Curve	50°C Hot Curve	–6°C (Still Air) Cold Curve	50°C Hot Curve
25	0.08	0.11	0.03	0.08	0.04	0.08
50	0.31	0.45	0.12	0.32	0.15	0.33
75	0.70	1.02	0.28	0.72	0.34	0.74
100	1.24	1.81	0.49	1.28	0.60	1.32
125	1.94	2.83	0.77	2.00	0.94	2.07
150	2.79	4.08	1.11	2.88	1.35	2.98
175	3.80	5.55	1.51	3.92	1.84	4.05
200	4.96	7.25	1.98	5.12	2.40	5.29
250	7.75	11.32	3.09	8.00	3.75	8.27
300	11.16	16.31	4.45	11.52	5.40	11.90
350	15.19	22.20	6.06	15.69	7.35	16.20
400	19.85	28.99	7.91	20.49	9.59	21.16
450	25.12	36.69	10.01	25.93	12.14	26.78
500	31.01	45.30	12.36	32.01	14.99	33.06
550	37.52	54.81	14.96	38.73	18.14	40.01
600	44.65	65.23	17.80	46.10	21.59	47.61

Table 10 – Ordinates for Plotting Uplift and Ground Clearance Templates – PVC-covered Conductors

1. 16 mm² Copper and 25 mm² Aluminium Alloy and ACSR (Basic Span 90 m)

Span (m)	Sag (m)					
	16 mm ² Cu		25 mm ² Al. Alloy		25 mm ² ACSR	
	-6°C (Still Air) Cold Curve	50°C Hot Curve	-6°C (Still Air) Cold Curve	50°C Hot Curve	-6°C (Still Air) Cold Curve	50°C Hot Curve
25	0.17	0.21	0.11	0.17	0.10	0.16
50	0.69	0.85	0.43	0.70	0.38	0.62
75	1.55	1.92	0.97	1.57	0.86	1.40
100	2.76	3.41	1.73	2.79	1.54	2.49
125	4.31	5.32	2.70	4.35	2.40	3.89
150	6.21	7.67	3.89	6.27	3.46	5.60
175	8.45	10.43	5.30	8.53	4.71	7.63
200	11.04	13.63	6.92	11.14	6.15	9.96
250	17.25	21.29	10.81	17.41	9.61	15.57
300	24.84	30.66	15.56	25.07	13.84	22.42
350	33.81	41.74	21.18	34.13	18.83	30.52
400	44.17	54.51	27.67	44.57	24.60	39.86
450	55.90	68.99	35.02	56.41	31.13	50.44
500	69.01	85.18	43.23	69.65	38.43	62.28
550	83.50	103.07	52.31	84.27	46.51	75.35
600	99.37	122.66	62.25	100.29	55.35	89.68

Table 10 (Cont'd)

2. 32 mm² Copper and 50 mm² Aluminium Alloy and ACSR (Basic Span 125 m)

Span (m)	Sag (m)					
	32 mm ² Cu		50 mm ² Al. Alloy		50 mm ² ACSR	
	–6°C (Still Air) Cold Curve	50°C Hot Curve	–6°C (Still Air) Cold Curve	50°C Hot Curve	–6°C (Still Air) Cold Curve	50°C Hot Curve
25	0.12	0.15	0.07	0.12	0.07	0.11
50	0.49	0.60	0.28	0.47	0.29	0.45
75	1.10	1.36	0.62	1.05	0.65	1.02
100	1.95	2.41	1.11	1.86	1.16	1.82
125	3.04	3.76	1.73	2.91	1.81	2.84
150	4.38	5.42	2.49	4.19	2.61	4.08
175	5.97	7.38	3.39	5.70	3.55	5.56
200	7.79	9.64	4.43	7.45	4.64	7.26
250	12.17	15.06	6.93	11.64	7.25	11.34
300	17.53	21.68	9.97	16.76	10.44	16.34
350	23.86	29.51	13.58	22.81	14.22	22.24
400	31.17	38.55	17.73	29.79	18.57	29.04
450	39.44	48.79	22.44	37.71	23.50	36.76
500	48.70	60.23	27.71	46.55	29.01	45.38
550	58.92	72.88	33.52	56.33	35.10	54.91
600	70.12	86.73	39.90	67.04	41.78	65.35

Design Sag/Tension Tables for 16 mm² Copper Conductor — Basic Span 90 m (Tables 11 & 12)

Table 11 — Bare Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)								
		30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0
— 6.0	137	0.12	0.22	0.34	0.48	0.66	0.86	1.09	1.35	1.63
0.0	128	0.13	0.23	0.36	0.52	0.71	0.92	1.17	1.44	1.75
5.0	121	0.14	0.24	0.38	0.55	0.75	0.98	1.23	1.52	1.84
10.0	115	0.14	0.26	0.40	0.58	0.79	1.03	1.30	1.61	1.94
15.0	109	0.15	0.27	0.42	0.61	0.83	1.08	1.37	1.69	2.04
20.0	104	0.16	0.28	0.44	0.64	0.87	1.13	1.43	1.77	2.14
25.0	100	0.17	0.30	0.46	0.67	0.91	1.18	1.50	1.85	2.24
30.0	96	0.17	0.31	0.48	0.69	0.95	1.24	1.56	1.93	2.34
35.0	92	0.18	0.32	0.50	0.72	0.98	1.29	1.63	2.01	2.43
40.0	88	0.19	0.33	0.52	0.75	1.02	1.34	1.69	2.09	2.52
45.0	85	0.19	0.35	0.54	0.78	1.06	1.38	1.75	2.16	2.62
50.0	82	0.20	0.36	0.56	0.81	1.10	1.43	1.81	2.24	2.71

Table 12 — PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)								
		30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0
— 6.0	99	0.25	0.44	0.69	0.99	1.35	1.77	2.24	2.76	3.34
0.0	97	0.26	0.45	0.71	1.02	1.39	1.81	2.30	2.83	3.43
5.0	95	0.26	0.46	0.72	1.04	1.42	1.85	2.35	2.90	3.50
10.0	93	0.27	0.47	0.74	1.06	1.45	1.89	2.39	2.96	3.58
15.0	91	0.27	0.48	0.75	1.09	1.48	1.93	2.44	3.01	3.65
20.0	89	0.28	0.49	0.77	1.11	1.51	1.97	2.49	3.07	3.72
25.0	87	0.28	0.50	0.78	1.13	1.53	2.00	2.54	3.13	3.79
30.0	86	0.29	0.51	0.80	1.15	1.56	2.04	2.58	3.19	3.86
35.0	84	0.29	0.52	0.81	1.17	1.59	2.08	2.63	3.24	3.92
40.0	83	0.30	0.53	0.82	1.19	1.62	2.11	2.67	3.30	3.99
45.0	82	0.30	0.54	0.84	1.21	1.64	2.15	2.72	3.35	4.06
50.0	80	0.31	0.55	0.85	1.23	1.67	2.18	2.76	3.41	4.12

Design Sag/Tension Tables for 32 mm² Copper Conductor – Basic Span 125 m (Tables 13 & 14)

Table 13 – Bare Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)									
		60.0	70.0	80.0	90.0	100.0	110.0	120.0	125.0	130.0	140.0
– 6.0	299	0.45	0.61	0.79	1.00	1.24	1.50	1.79	1.94	2.10	2.43
0.0	284	0.47	0.64	0.83	1.06	1.30	1.58	1.88	2.04	2.20	2.56
5.0	273	0.49	0.66	0.87	1.10	1.36	1.64	1.95	2.12	2.29	2.66
10.0	263	0.51	0.69	0.90	1.14	1.41	1.71	2.03	2.20	2.38	2.76
15.0	253	0.53	0.72	0.94	1.18	1.46	1.77	2.10	2.28	2.47	2.86
20.0	245	0.54	0.74	0.97	1.23	1.51	1.83	2.18	2.36	2.56	2.97
25.0	237	0.56	0.77	1.00	1.27	1.56	1.89	2.25	2.44	2.64	3.07
30.0	229	0.58	0.79	1.03	1.31	1.62	1.95	2.33	2.52	2.73	3.17
35.0	222	0.60	0.82	1.07	1.35	1.67	2.02	2.40	2.60	2.81	3.26
40.0	216	0.62	0.84	1.10	1.39	1.72	2.08	2.47	2.68	2.90	3.36
45.0	210	0.63	0.86	1.13	1.43	1.76	2.13	2.54	2.76	2.98	3.46
50.0	204	0.65	0.89	1.16	1.47	1.81	2.19	2.61	2.83	3.06	3.55

Table 14 – PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)									
		60.0	70.0	80.0	90.0	100.0	110.0	120.0	125.0	130.0	140.0
– 6.0	250	0.70	0.95	1.25	1.58	1.95	2.36	2.80	3.04	3.29	3.82
0.0	243	0.72	0.98	1.28	1.62	2.00	2.42	2.88	3.13	3.38	3.92
5.0	238	0.74	1.00	1.31	1.66	2.04	2.47	2.94	3.19	3.45	4.01
10.0	233	0.75	1.02	1.34	1.69	2.09	2.52	3.00	3.26	3.53	4.09
15.0	229	0.77	1.04	1.36	1.72	2.13	2.58	3.06	3.33	3.60	4.17
20.0	224	0.78	1.06	1.39	1.76	2.17	2.63	3.12	3.39	3.67	4.25
25.0	220	0.80	1.08	1.42	1.79	2.21	2.68	3.18	3.45	3.74	4.33
30.0	216	0.81	1.10	1.44	1.82	2.25	2.72	3.24	3.52	3.81	4.41
35.0	212	0.83	1.12	1.47	1.86	2.29	2.77	3.30	3.58	3.87	4.49
40.0	209	0.84	1.14	1.49	1.89	2.33	2.82	3.36	3.64	3.94	4.57
45.0	205	0.85	1.16	1.52	1.92	2.37	2.87	3.41	3.70	4.01	4.65
50.0	202	0.87	1.18	1.54	1.95	2.41	2.92	3.47	3.76	4.07	4.72

Design Sag/Tension Tables for 25 mm² Aluminium Alloy Conductor – Basic Span 90 m (Tables 15 & 16)

Table 15 – Bare Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)								
		30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0
– 6.0	198	0.05	0.08	0.13	0.19	0.25	0.33	0.42	0.52	0.62
0.0	176	0.05	0.09	0.15	0.21	0.28	0.37	0.47	0.58	0.70
5.0	158	0.06	0.10	0.16	0.23	0.32	0.41	0.52	0.65	0.78
10.0	142	0.06	0.12	0.18	0.26	0.35	0.46	0.58	0.72	0.87
15.0	126	0.07	0.13	0.20	0.29	0.40	0.52	0.66	0.81	0.98
20.0	112	0.08	0.15	0.23	0.33	0.45	0.58	0.74	0.91	1.10
25.0	100	0.09	0.16	0.26	0.37	0.50	0.66	0.83	1.02	1.24
30.0	89	0.10	0.18	0.29	0.41	0.56	0.73	0.93	1.14	1.39
35.0	80	0.11	0.20	0.32	0.46	0.62	0.81	1.03	1.27	1.54
40.0	73	0.13	0.22	0.35	0.50	0.69	0.90	1.13	1.40	1.69
45.0	67	0.14	0.24	0.38	0.55	0.75	0.98	1.24	1.53	1.85
50.0	62	0.15	0.26	0.41	0.60	0.81	1.06	1.34	1.65	2.00

Table 16 – PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)								
		30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0
– 6.0	122	0.16	0.28	0.43	0.62	0.85	1.11	1.40	1.73	2.09
0.0	114	0.17	0.30	0.46	0.67	0.91	1.18	1.50	1.85	2.24
5.0	108	0.18	0.31	0.49	0.70	0.96	1.25	1.58	1.95	2.36
10.0	103	0.18	0.33	0.51	0.74	1.00	1.31	1.66	2.05	2.48
15.0	98	0.19	0.34	0.54	0.77	1.05	1.37	1.74	2.15	2.60
20.0	94	0.20	0.36	0.56	0.81	1.10	1.44	1.82	2.24	2.72
25.0	90	0.21	0.37	0.58	0.84	1.15	1.50	1.89	2.34	2.83
30.0	87	0.22	0.39	0.61	0.88	1.19	1.56	1.97	2.43	2.94
35.0	84	0.23	0.40	0.63	0.91	1.24	1.61	2.04	2.52	3.05
40.0	81	0.24	0.42	0.65	0.94	1.28	1.67	2.12	2.61	3.16
45.0	78	0.24	0.43	0.68	0.97	1.32	1.73	2.19	2.70	3.27
50.0	76	0.25	0.45	0.70	1.00	1.37	1.78	2.26	2.79	3.37

Design Sag/Tension Tables for 50 mm² Aluminium Alloy Conductor – Basic Span 125 m (Tables 17 & 18)

Table 17 – Bare Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)									
		60.0	70.0	80.0	90.0	100.0	110.0	120.0	125.0	130.0	140.0
– 6.0	414	0.18	0.24	0.32	0.40	0.49	0.60	0.71	0.77	0.84	0.97
0.0	373	0.20	0.27	0.35	0.45	0.55	0.66	0.79	0.86	0.93	1.08
5.0	340	0.22	0.29	0.39	0.49	0.60	0.73	0.87	0.94	1.02	1.18
10.0	310	0.24	0.32	0.42	0.53	0.66	0.80	0.95	1.03	1.12	1.29
15.0	282	0.26	0.36	0.46	0.59	0.73	0.88	1.04	1.13	1.23	1.42
20.0	257	0.29	0.39	0.51	0.65	0.80	0.96	1.15	1.24	1.35	1.56
25.0	235	0.31	0.43	0.56	0.71	0.87	1.06	1.26	1.36	1.47	1.71
30.0	215	0.34	0.47	0.61	0.77	0.95	1.15	1.37	1.49	1.61	1.87
35.0	198	0.37	0.51	0.66	0.84	1.03	1.25	1.49	1.62	1.75	2.03
40.0	183	0.40	0.55	0.71	0.90	1.12	1.35	1.61	1.74	1.89	2.19
45.0	170	0.43	0.59	0.77	0.97	1.20	1.45	1.73	1.87	2.03	2.35
50.0	160	0.46	0.63	0.82	1.04	1.28	1.55	1.85	2.00	2.17	2.51

Table 18 – PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)									
		60.0	70.0	80.0	90.0	100.0	110.0	120.0	125.0	130.0	140.0
– 6.0	315	0.40	0.54	0.71	0.90	1.11	1.34	1.60	1.73	1.87	2.17
0.0	294	0.43	0.58	0.76	0.96	1.19	1.44	1.71	1.86	2.01	2.33
5.0	278	0.45	0.62	0.81	1.02	1.26	1.52	1.81	1.97	2.13	2.47
10.0	263	0.48	0.65	0.85	1.08	1.33	1.61	1.91	2.07	2.24	2.60
15.0	250	0.50	0.68	0.89	1.13	1.40	1.69	2.01	2.18	2.36	2.74
20.0	238	0.53	0.72	0.94	1.19	1.47	1.77	2.11	2.29	2.48	2.87
25.0	228	0.55	0.75	0.98	1.24	1.53	1.86	2.21	2.40	2.59	3.01
30.0	218	0.58	0.79	1.03	1.30	1.60	1.94	2.31	2.50	2.71	3.14
35.0	209	0.60	0.82	1.07	1.35	1.67	2.02	2.40	2.61	2.82	3.27
40.0	201	0.62	0.85	1.11	1.40	1.73	2.10	2.50	2.71	2.93	3.40
45.0	194	0.65	0.88	1.15	1.46	1.80	2.18	2.59	2.81	3.04	3.53
50.0	188	0.67	0.91	1.19	1.51	1.86	2.25	2.68	2.91	3.15	3.65

Design Sag/Tension Tables for 25 mm² ACSR Conductor – Basic Span 90 m (Tables 19 & 20)

Table 19 – Bare Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)								
		30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0
– 6.0	239	0.05	0.09	0.14	0.20	0.27	0.35	0.45	0.55	0.67
0.0	214	0.06	0.10	0.15	0.22	0.30	0.39	0.50	0.62	0.75
5.0	195	0.06	0.11	0.17	0.24	0.33	0.43	0.55	0.68	0.82
10.0	177	0.07	0.12	0.19	0.27	0.37	0.48	0.61	0.75	0.90
15.0	160	0.07	0.13	0.21	0.30	0.41	0.53	0.67	0.83	1.00
20.0	144	0.08	0.15	0.23	0.33	0.45	0.59	0.74	0.91	1.11
25.0	131	0.09	0.16	0.25	0.36	0.50	0.65	0.82	1.01	1.22
30.0	119	0.10	0.18	0.28	0.40	0.55	0.71	0.90	1.11	1.35
35.0	108	0.11	0.19	0.30	0.44	0.60	0.78	0.99	1.22	1.47
40.0	99	0.12	0.21	0.33	0.48	0.65	0.85	1.07	1.33	1.61
45.0	92	0.13	0.23	0.36	0.52	0.70	0.92	1.16	1.44	1.74
50.0	85	0.14	0.25	0.39	0.56	0.76	0.99	1.25	1.54	1.87

Table 20 – PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)								
		30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0	110.0
– 6.0	154	0.14	0.25	0.38	0.55	0.75	0.98	1.25	1.54	1.86
0.0	144	0.15	0.26	0.41	0.59	0.81	1.05	1.33	1.64	1.99
5.0	137	0.16	0.28	0.43	0.62	0.85	1.11	1.40	1.73	2.10
10.0	130	0.16	0.29	0.46	0.66	0.89	1.17	1.48	1.82	2.21
15.0	124	0.17	0.31	0.48	0.69	0.94	1.22	1.55	1.91	2.31
20.0	118	0.18	0.32	0.50	0.72	0.98	1.28	1.62	2.00	2.42
25.0	114	0.19	0.33	0.52	0.75	1.02	1.33	1.69	2.08	2.52
30.0	109	0.20	0.35	0.54	0.78	1.06	1.39	1.76	2.17	2.62
35.0	105	0.20	0.36	0.56	0.81	1.10	1.44	1.82	2.25	2.72
40.0	101	0.21	0.37	0.58	0.84	1.14	1.49	1.89	2.33	2.82
45.0	98	0.22	0.39	0.60	0.87	1.18	1.54	1.95	2.41	2.92
50.0	95	0.22	0.40	0.62	0.90	1.22	1.59	2.02	2.49	3.01

Design Sag/Tension Tables for 50 mm² ACSR Conductor – Basic Span 125 m (Tables 21 & 22)

Table 21 – Bare Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)									
		60.0	70.0	80.0	90.0	100.0	110.0	120.0	125.0	130.0	140.0
– 6.0	446	0.22	0.29	0.38	0.49	0.60	0.73	0.86	0.94	1.01	1.18
0.0	405	0.24	0.32	0.42	0.53	0.66	0.80	0.95	1.03	1.12	1.29
5.0	373	0.26	0.35	0.46	0.58	0.72	0.87	1.03	1.12	1.21	1.40
10.0	344	0.28	0.38	0.50	0.63	0.78	0.94	1.12	1.21	1.31	1.52
15.0	318	0.30	0.41	0.54	0.68	0.84	1.02	1.21	1.31	1.42	1.65
20.0	295	0.33	0.44	0.58	0.73	0.91	1.10	1.31	1.42	1.53	1.78
25.0	274	0.35	0.48	0.62	0.79	0.98	1.18	1.40	1.52	1.65	1.91
30.0	256	0.38	0.51	0.67	0.85	1.04	1.26	1.50	1.63	1.77	2.05
35.0	239	0.40	0.55	0.71	0.90	1.12	1.35	1.61	1.74	1.88	2.19
40.0	225	0.43	0.58	0.76	0.96	1.18	1.43	1.71	1.85	2.00	2.32
45.0	213	0.45	0.61	0.80	1.02	1.25	1.52	1.81	1.96	2.12	2.46
50.0	202	0.48	0.65	0.85	1.07	1.32	1.60	1.90	2.07	2.24	2.59

Table 22 – PVC-covered Conductor

Temp (°C)	Tension (kgf)	Sag (m) for span lengths (m)									
		60.0	70.0	80.0	90.0	100.0	110.0	120.0	125.0	130.0	140.0
– 6.0	355	0.42	0.57	0.74	0.94	1.16	1.40	1.67	1.81	1.96	2.27
0.0	334	0.44	0.60	0.79	1.00	1.23	1.49	1.78	1.93	2.08	2.42
5.0	318	0.47	0.63	0.83	1.05	1.29	1.57	1.86	2.02	2.19	2.54
10.0	304	0.49	0.66	0.87	1.10	1.35	1.64	1.95	2.12	2.29	2.65
15.0	291	0.51	0.69	0.91	1.15	1.41	1.71	2.04	2.21	2.39	2.77
20.0	279	0.53	0.72	0.94	1.19	1.47	1.78	2.12	2.30	2.49	2.89
25.0	269	0.55	0.75	0.98	1.24	1.53	1.86	2.21	2.40	2.59	3.00
30.0	259	0.57	0.78	1.02	1.29	1.59	1.93	2.29	2.49	2.69	3.12
35.0	250	0.59	0.81	1.06	1.34	1.65	1.99	2.37	2.58	2.79	3.23
40.0	241	0.61	0.84	1.09	1.38	1.71	2.06	2.46	2.66	2.88	3.34
45.0	234	0.63	0.86	1.13	1.43	1.76	2.13	2.54	2.75	2.98	3.45
50.0	227	0.65	0.89	1.16	1.47	1.82	2.20	2.61	2.84	3.07	3.56

**Table 23 – Maximum Allowable Weights at
Crossarm Ends**

Type of crossarm	Allowable weights (kg)
Intermediate and Pin Angle	385
Terminal	197
Section/Section angle for angles less than 60° (56°*)	321
Section/Section angle for angles greater than 60° (56°*)	Long end 321 Short end 500

Note: The above weights are assumed to be located at the positions of the outer phase conductors, and have been derived from maximum bending load consideration of the crossarm.

*56° for 50 mm² ACSR PVC-covered.

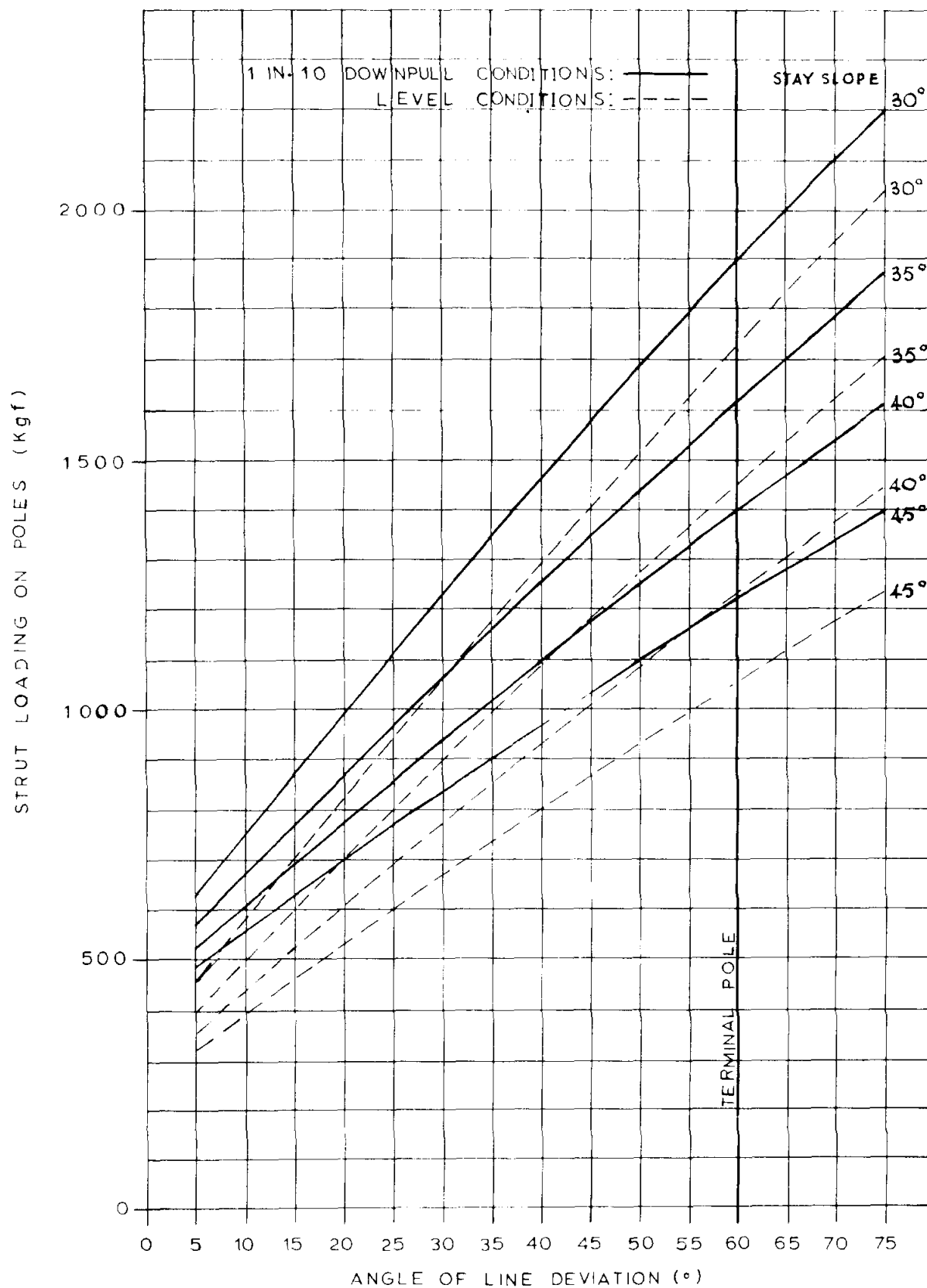


Chart 1 Strut loading of poles — 16 mm² bare copper

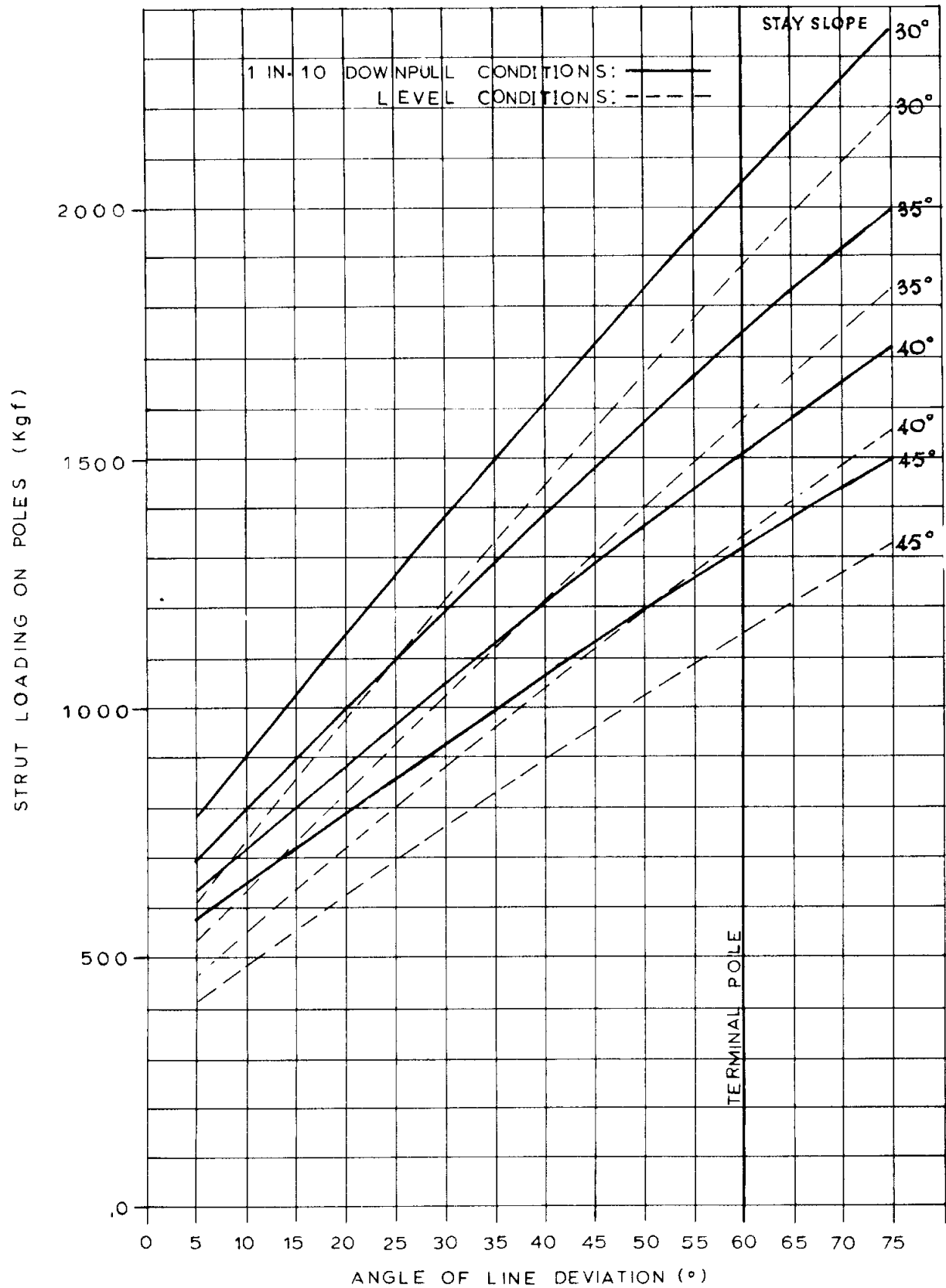


Chart 2 Strut loading of poles - 16 mm² copper PVC-covered

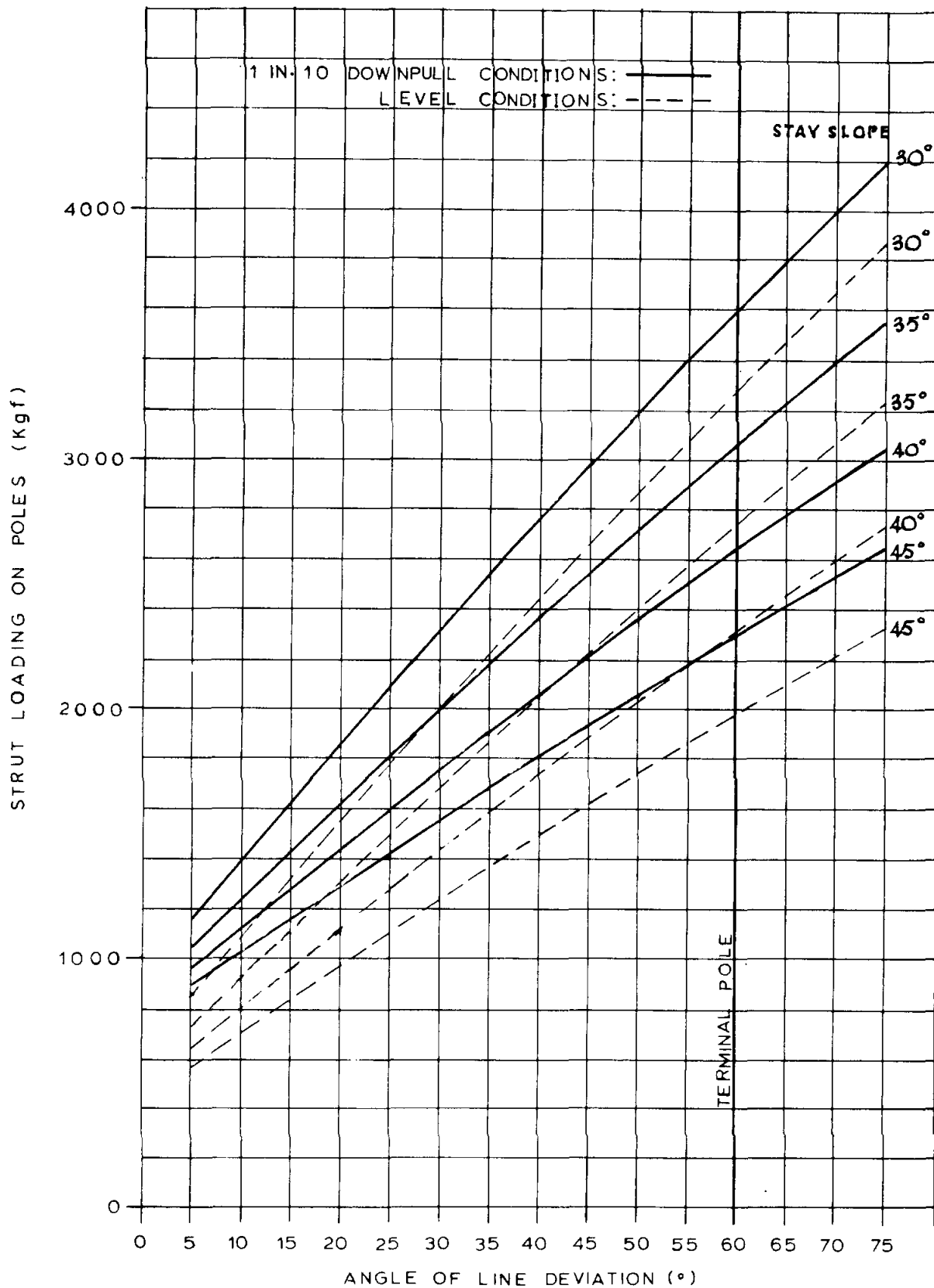


Chart 3 Strut loading of poles - 32 mm² bare copper

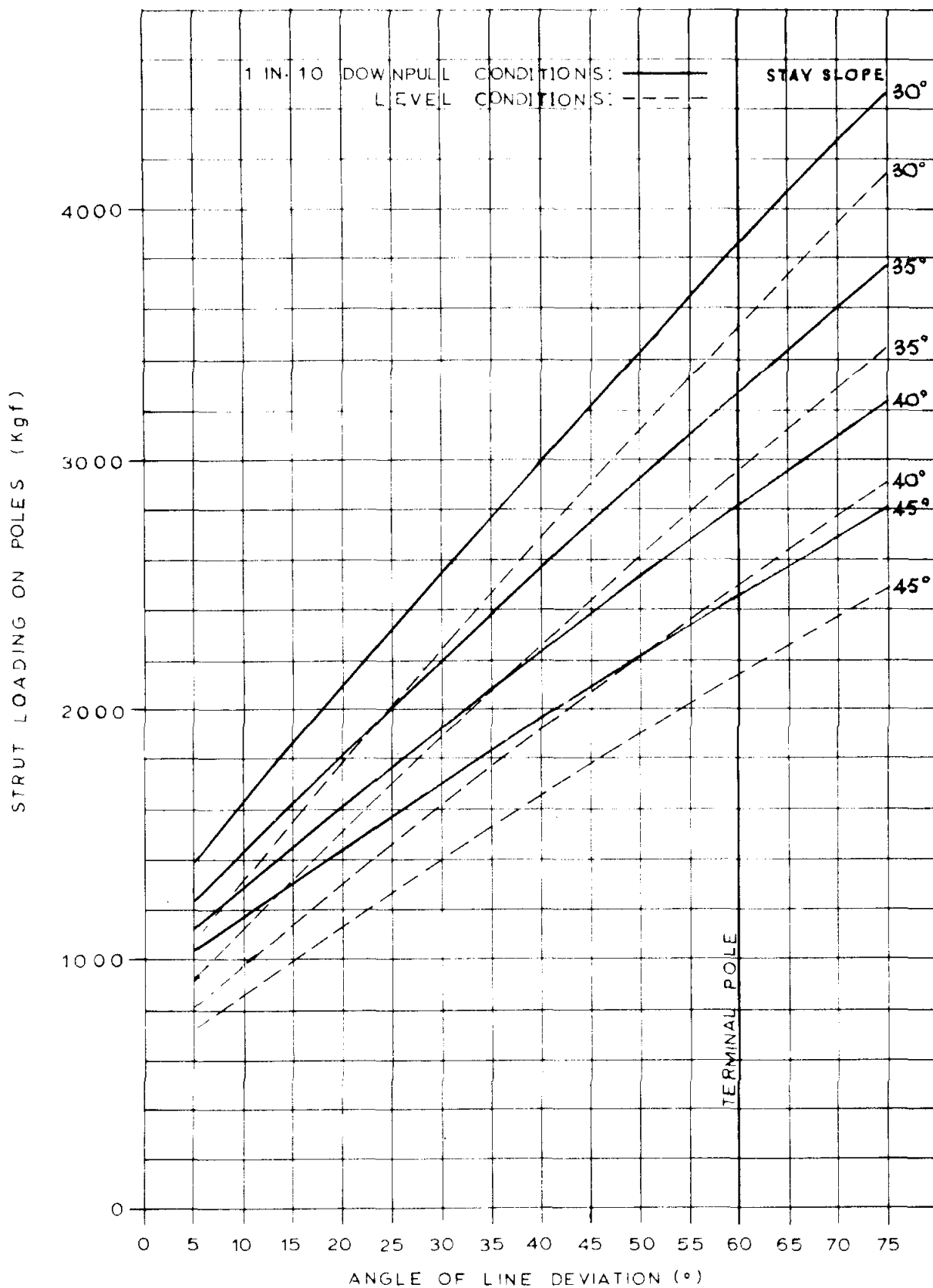


Chart 4 Strut loading of poles — 32 mm² copper PVC-covered

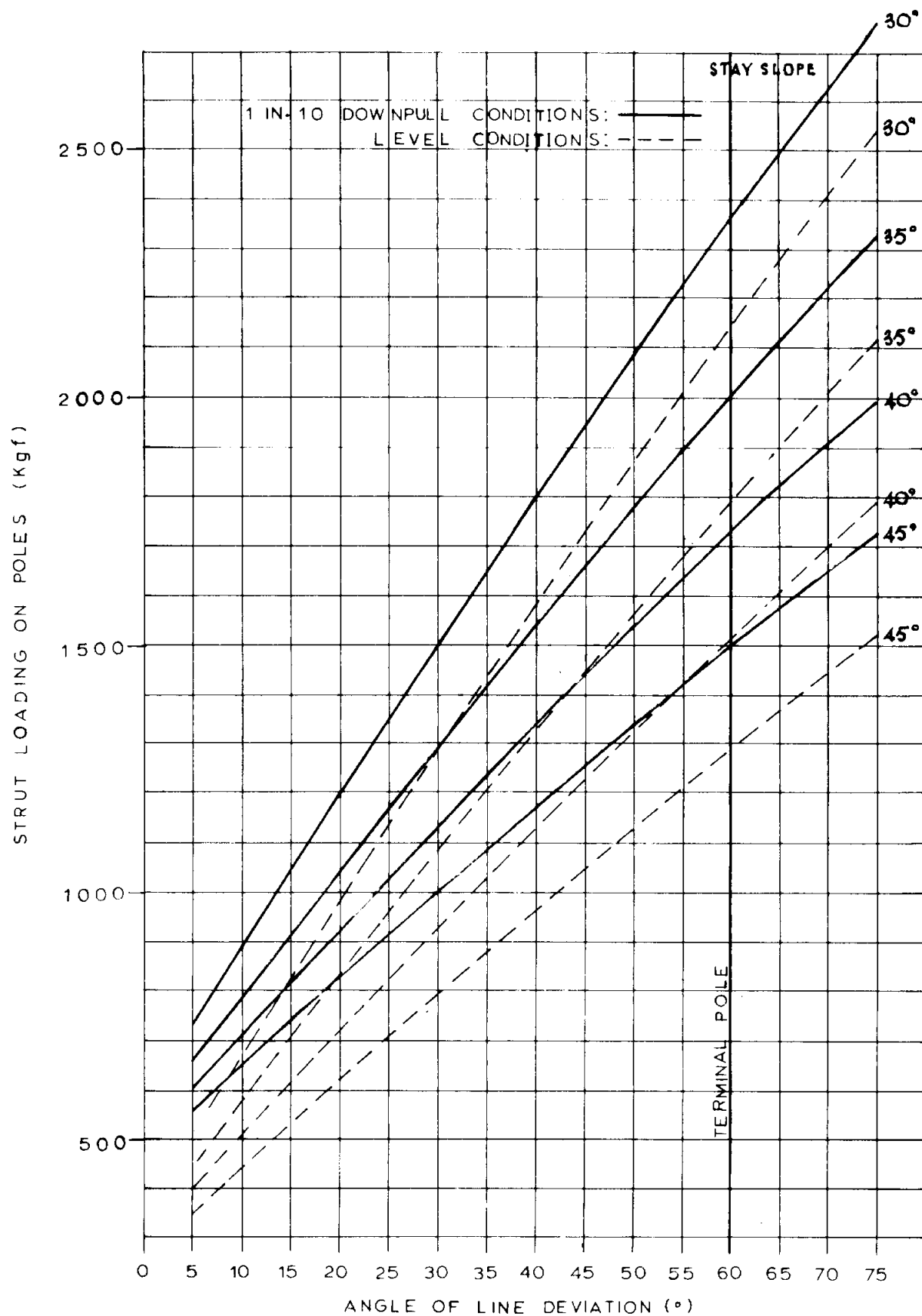


Chart 5 Strut loading of poles - 25 mm² bare aluminium alloy

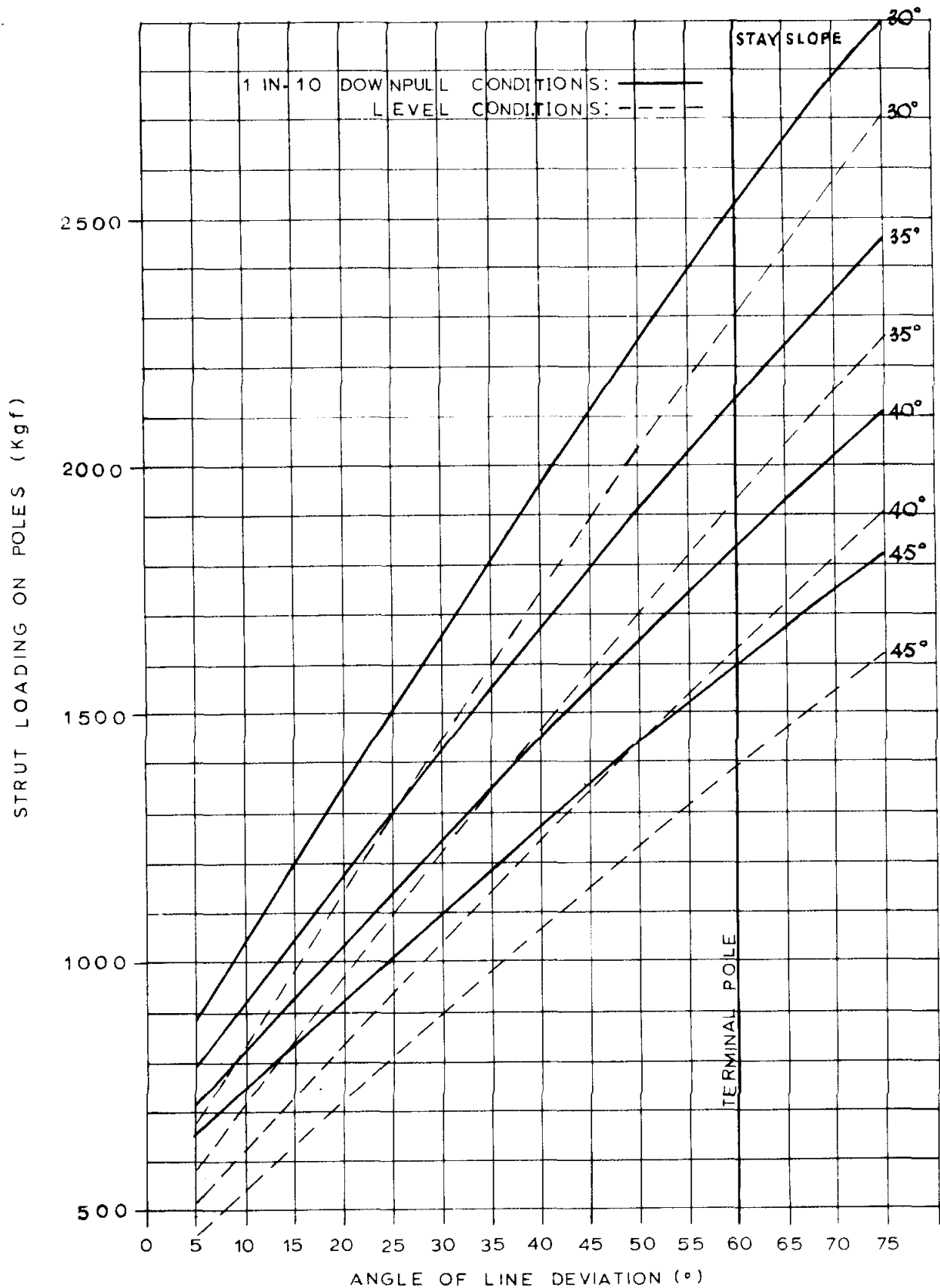


Chart 6 Strut loading of poles – 25 mm² aluminium alloy PVC-covered

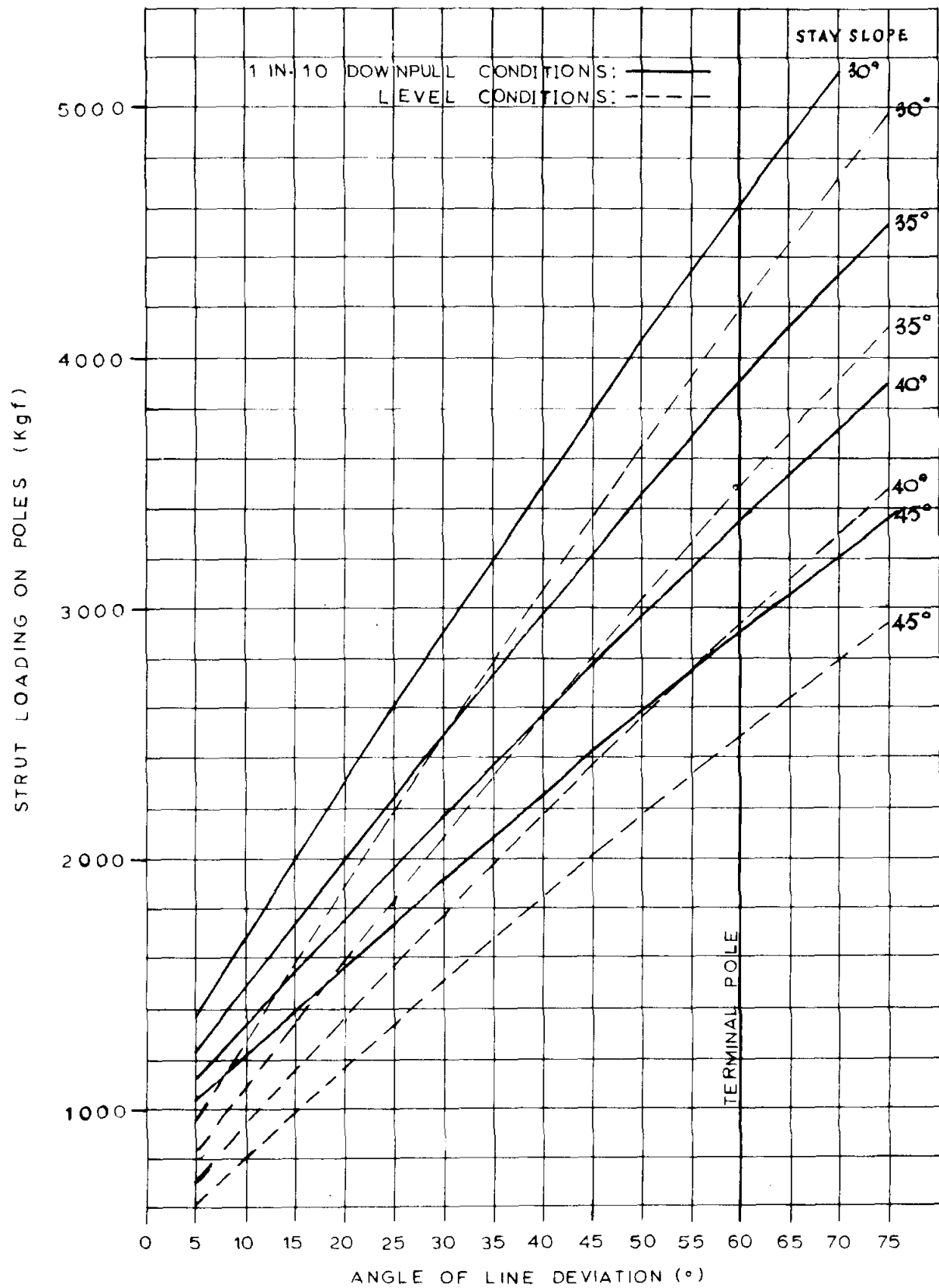


Chart 7 Strut loading of poles - 50 mm² bare aluminium alloy

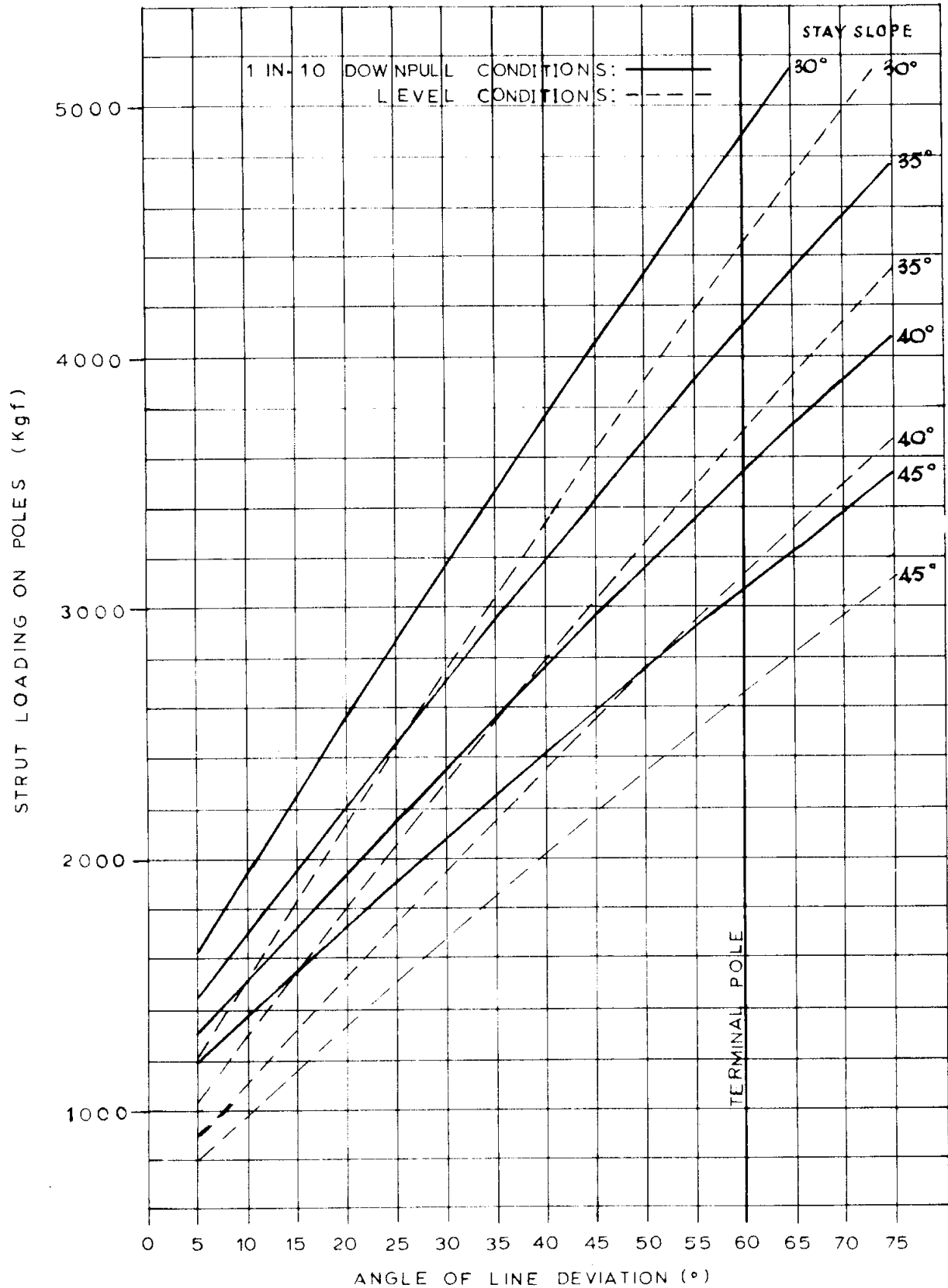


Chart 8 Strut loading of poles - 50 mm² aluminium alloy PVC-covered

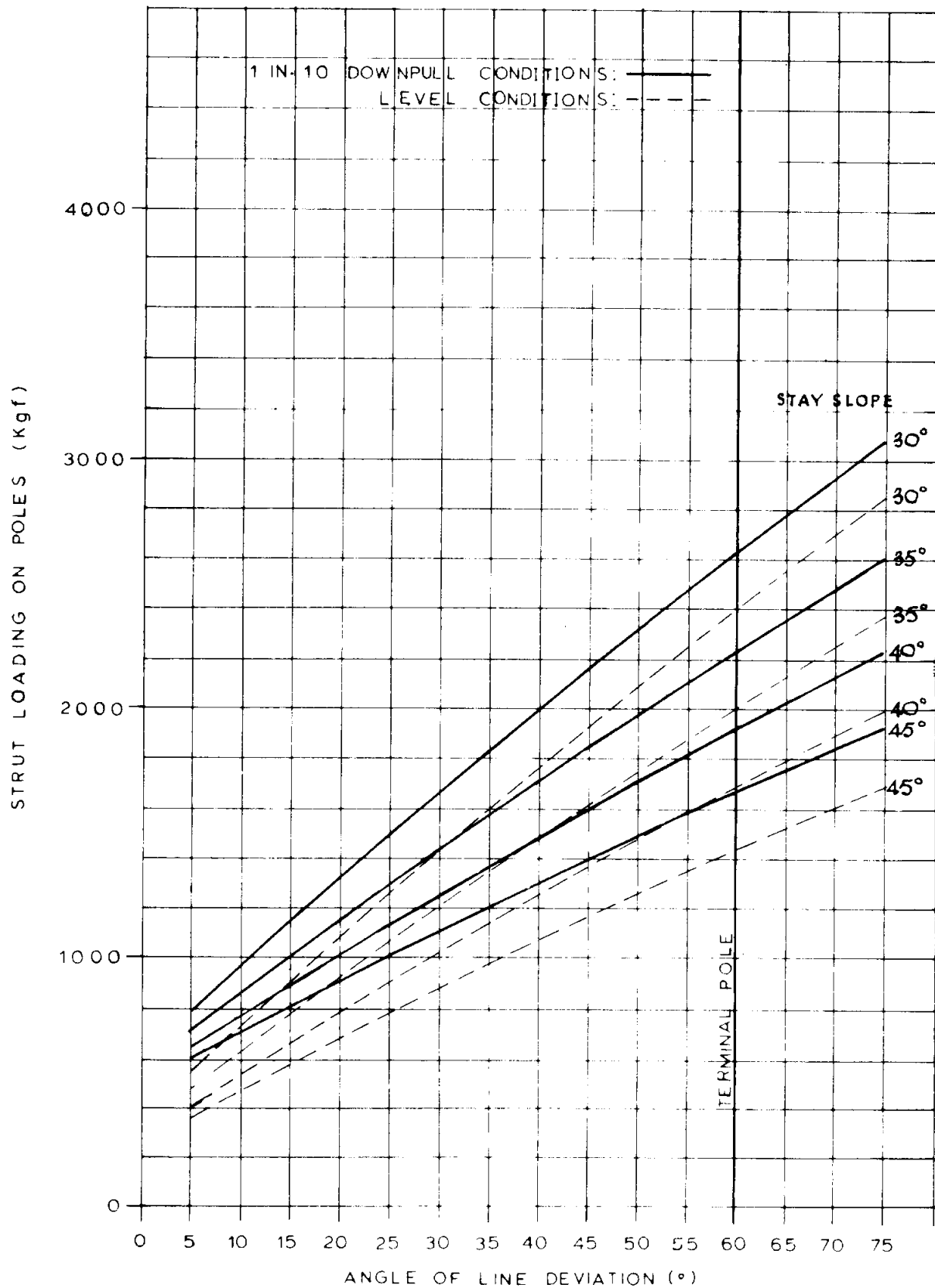


Chart 9 Strut loading of poles — 25 mm² bare ACSR

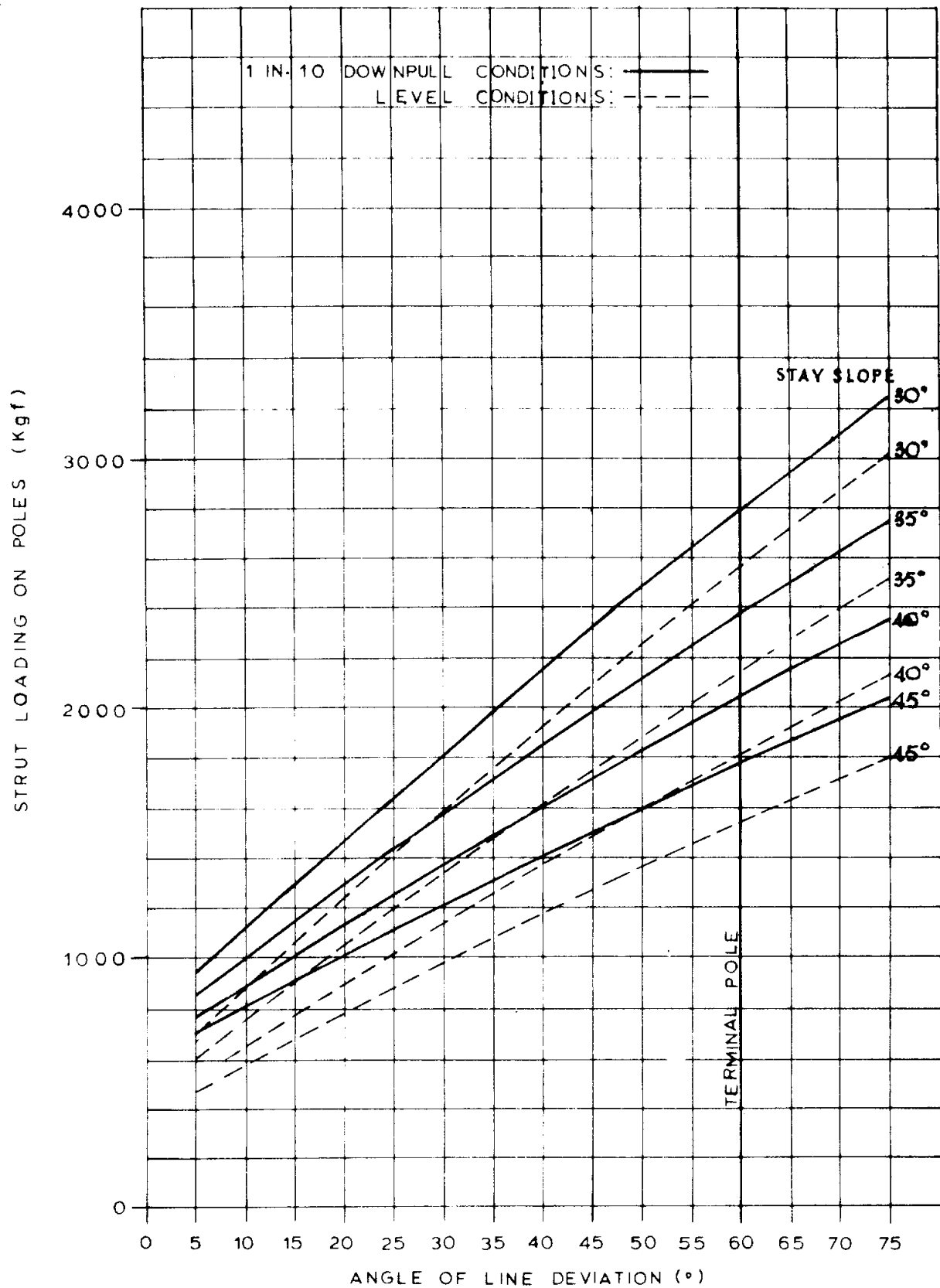


Chart 10 Strut loading of poles — 25 mm² ACSR PVC-covered

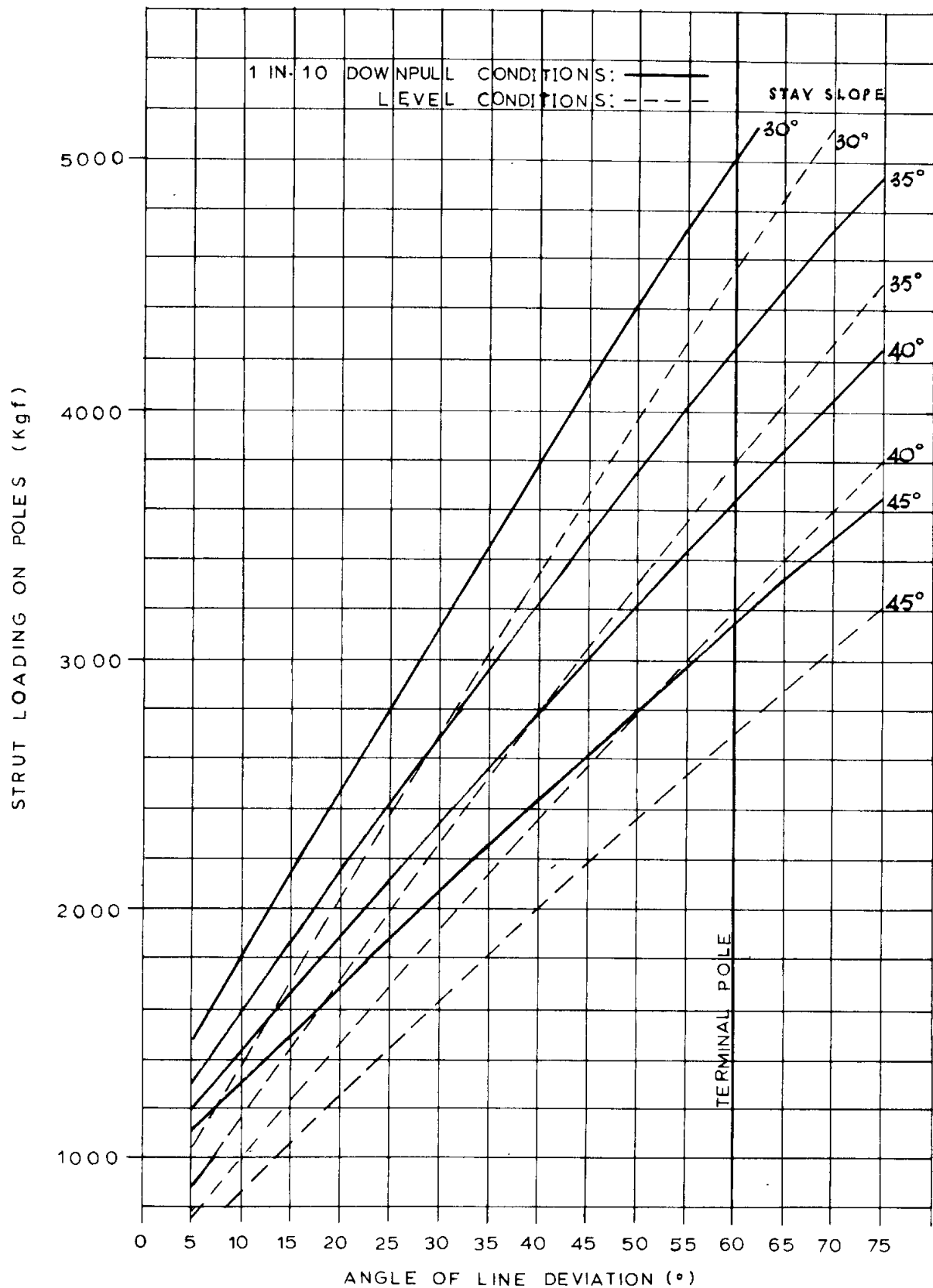


Chart 11 Strut loading of poles - 50 mm² bare ACSR

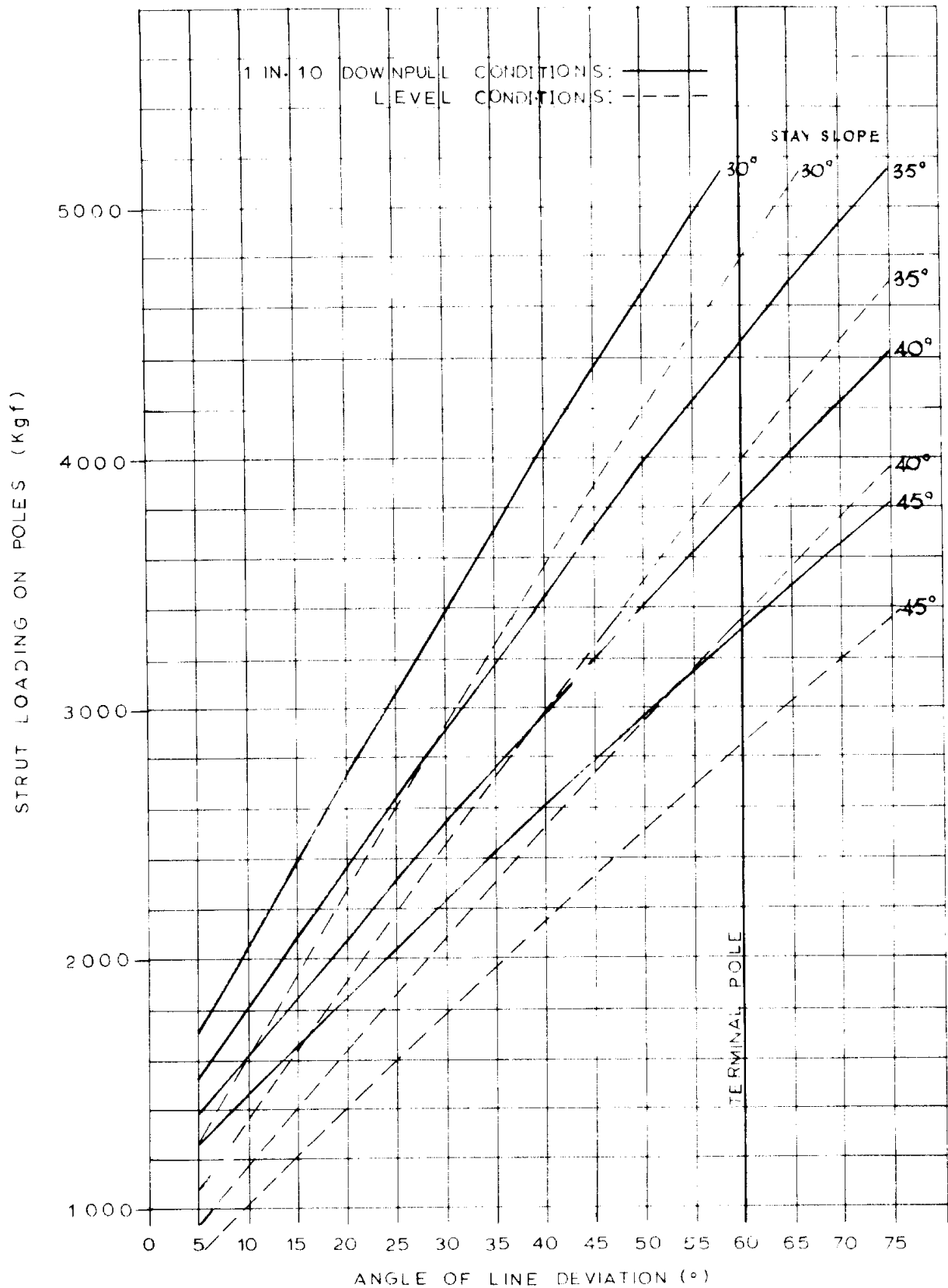
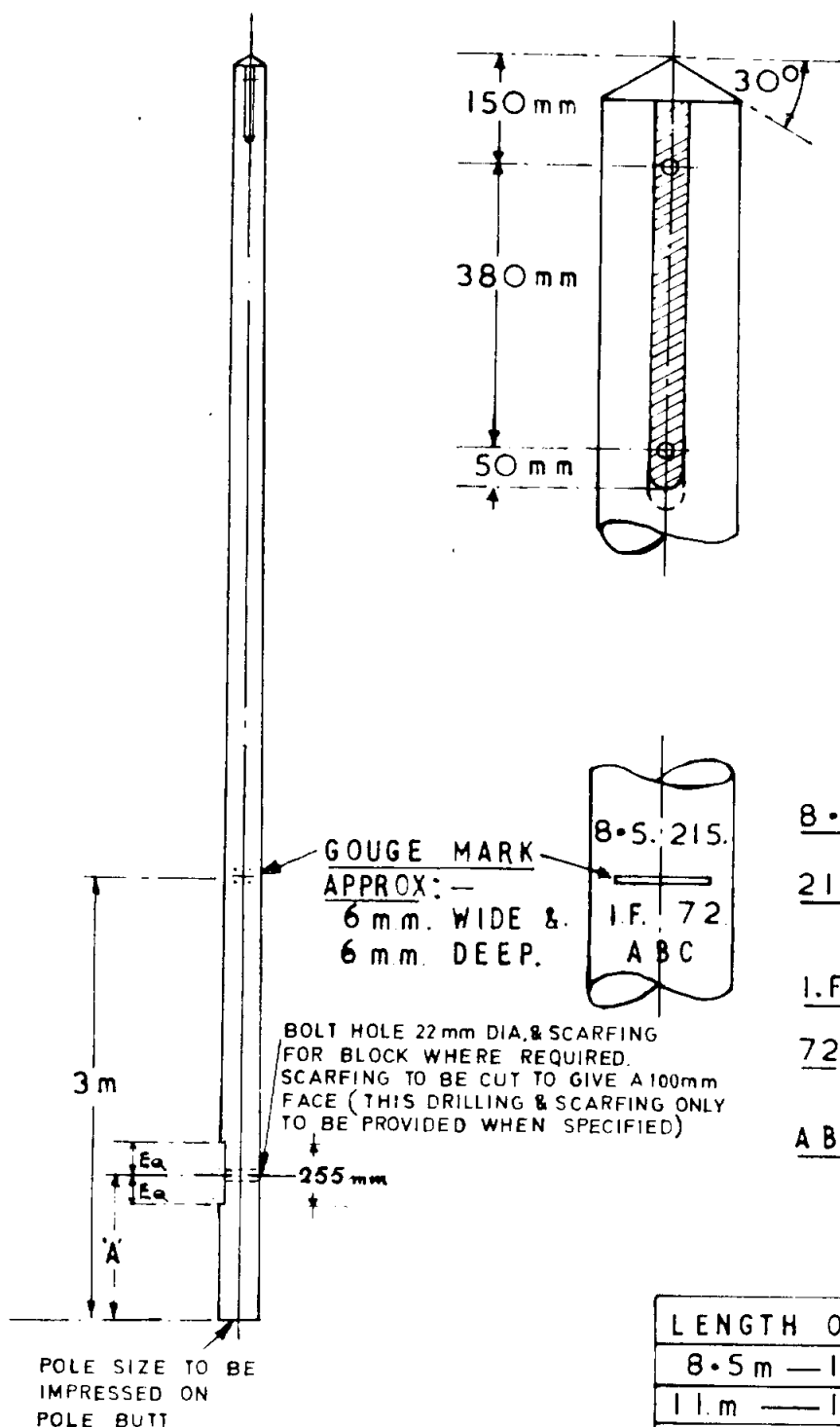


Chart 12 Strut loading of poles - 50 mm² ACSR PVC-covered



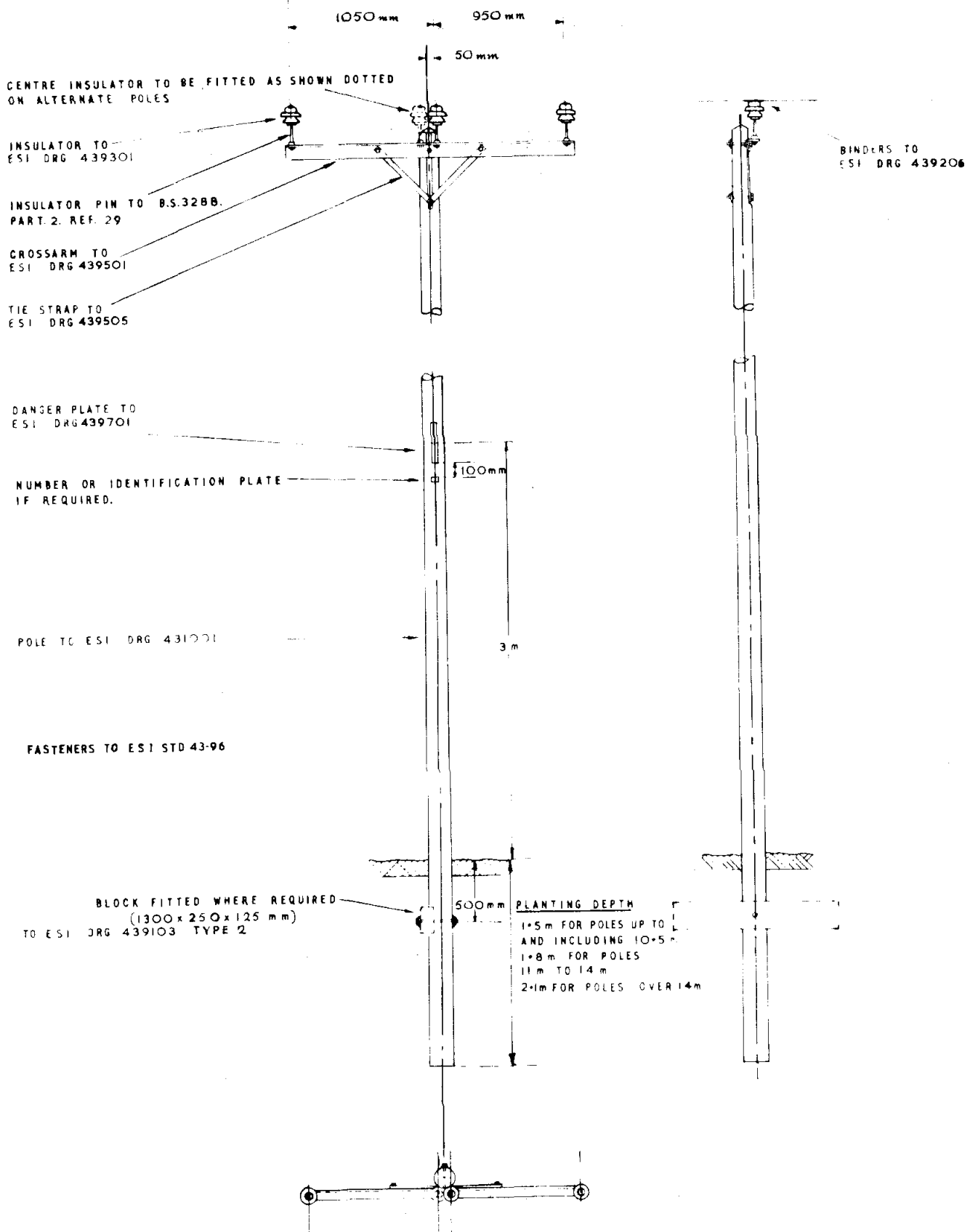
SCARFING TO BE CUT TO GIVE A 50 mm. FACE AND THE END CHAMFERED OFF BEYOND THE LENGTH SHOWN.
POLE SCARFED ON ONE SIDE ONLY.
DRILLINGS 22 mm. DIA.

REFERENCE.

8.5 = LENGTH OF POLE IN METRES.
215 = DIA OF POLE IN m.m. 1.5 m. FROM BUTT.
I.F. IMPORTED RED FIR.
72 YEAR OF CREOSOTING 1972.
ABC SUPPLIERS NAME OR MARK

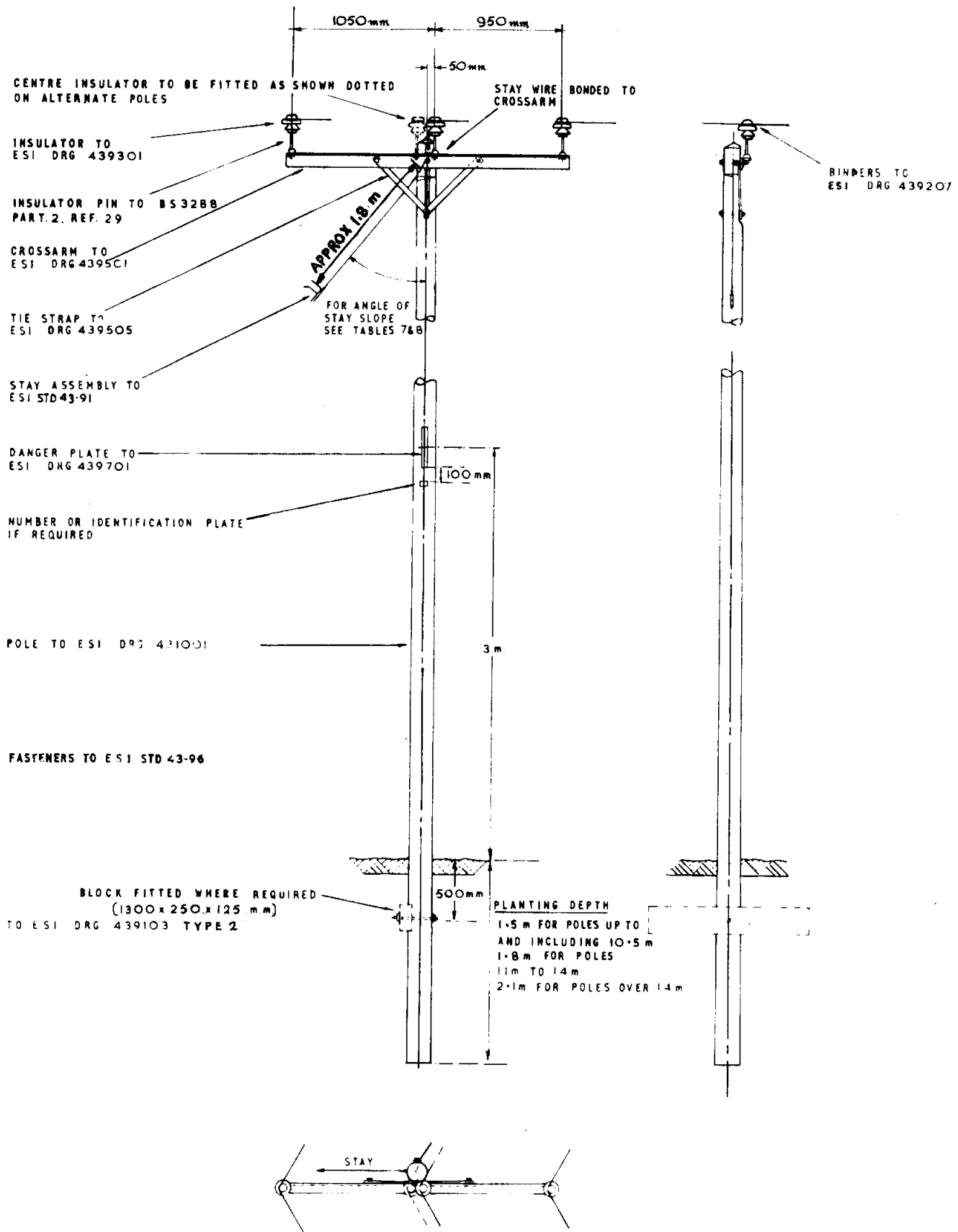
LENGTH OF POLE	DIMENSION 'A'
8.5 m — 10.5 m	1.0 m
11 m — 14 m	1.3 m
OVER 14 m	1.6 m

Fig. 1 Pole drilling, scarfing and marking



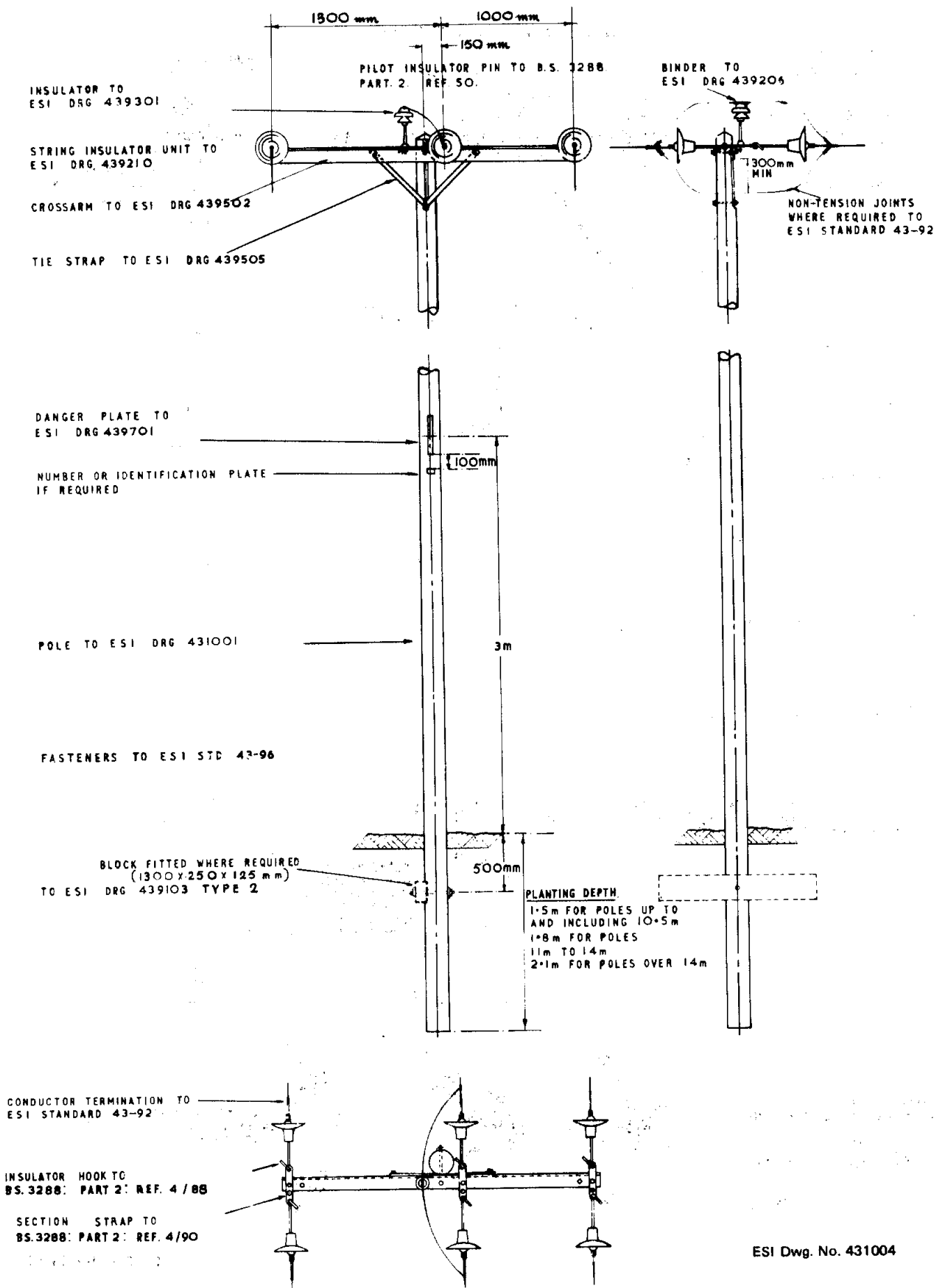
ESI Dwg. No. 431002

Fig. 2 General assembly - intermediate pole



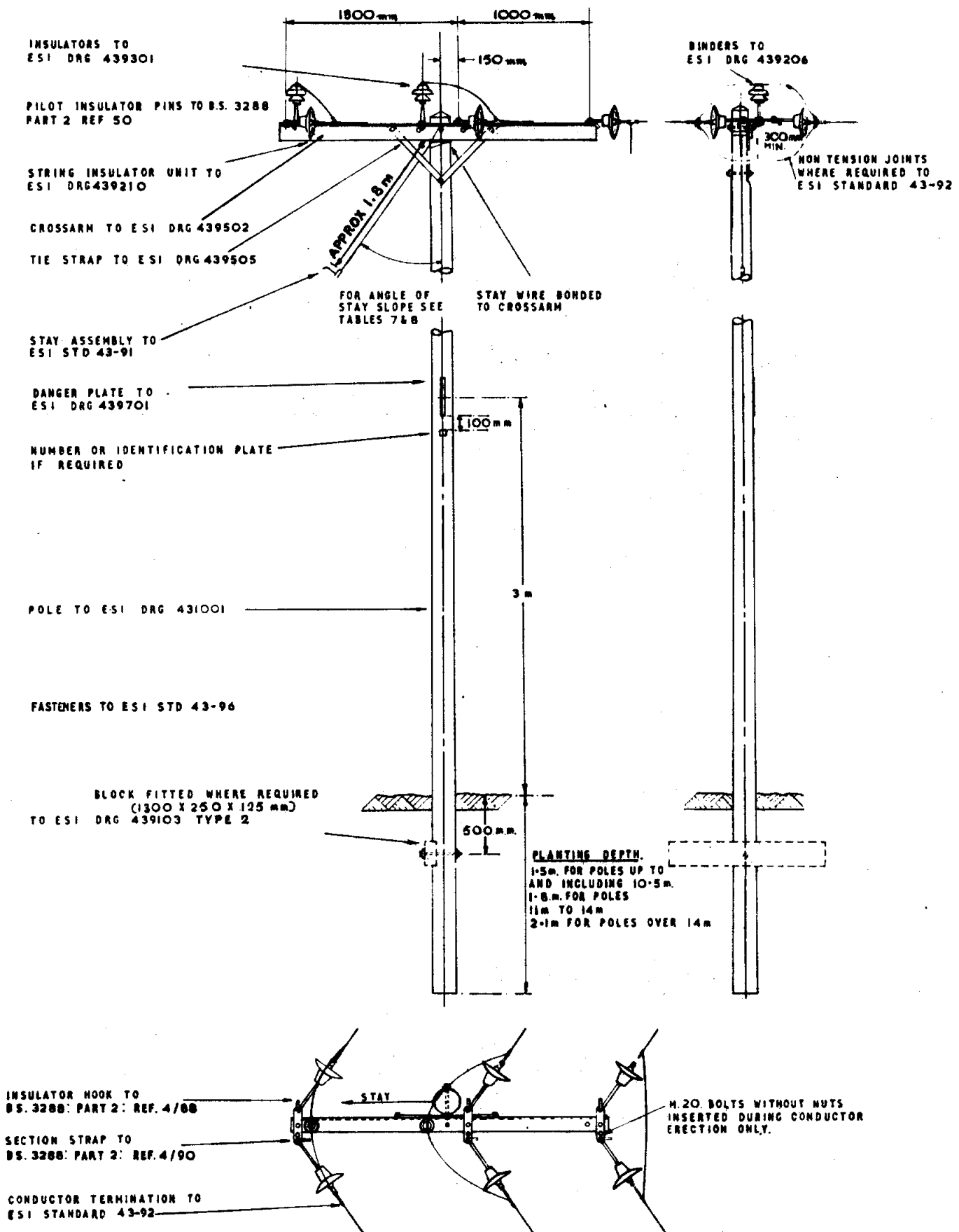
ESI Dwg. No. 431003

Fig. 3 General assembly — pin angle pole



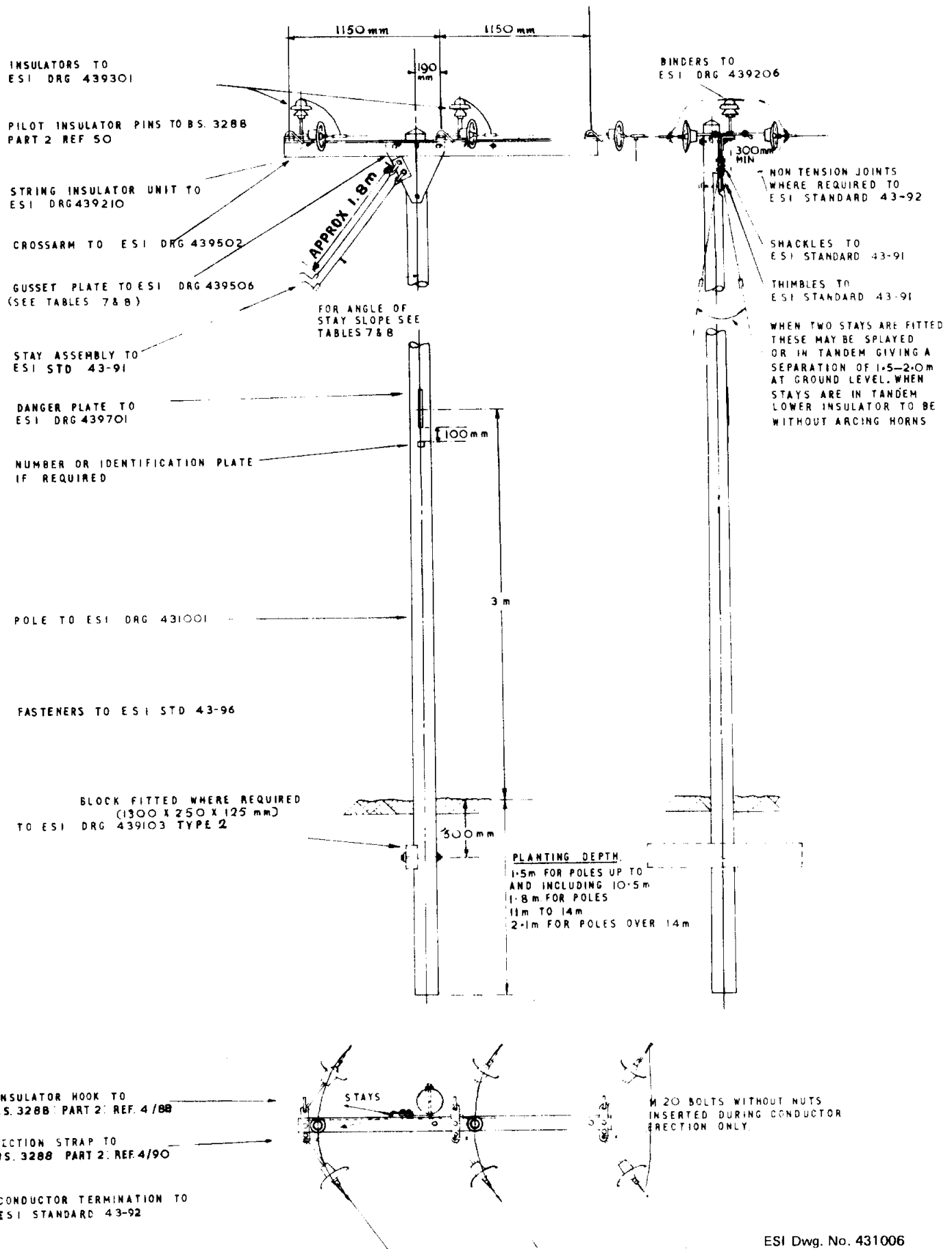
ESI Dwg. No. 431004

Fig. 4 General assembly -- straight line section pole



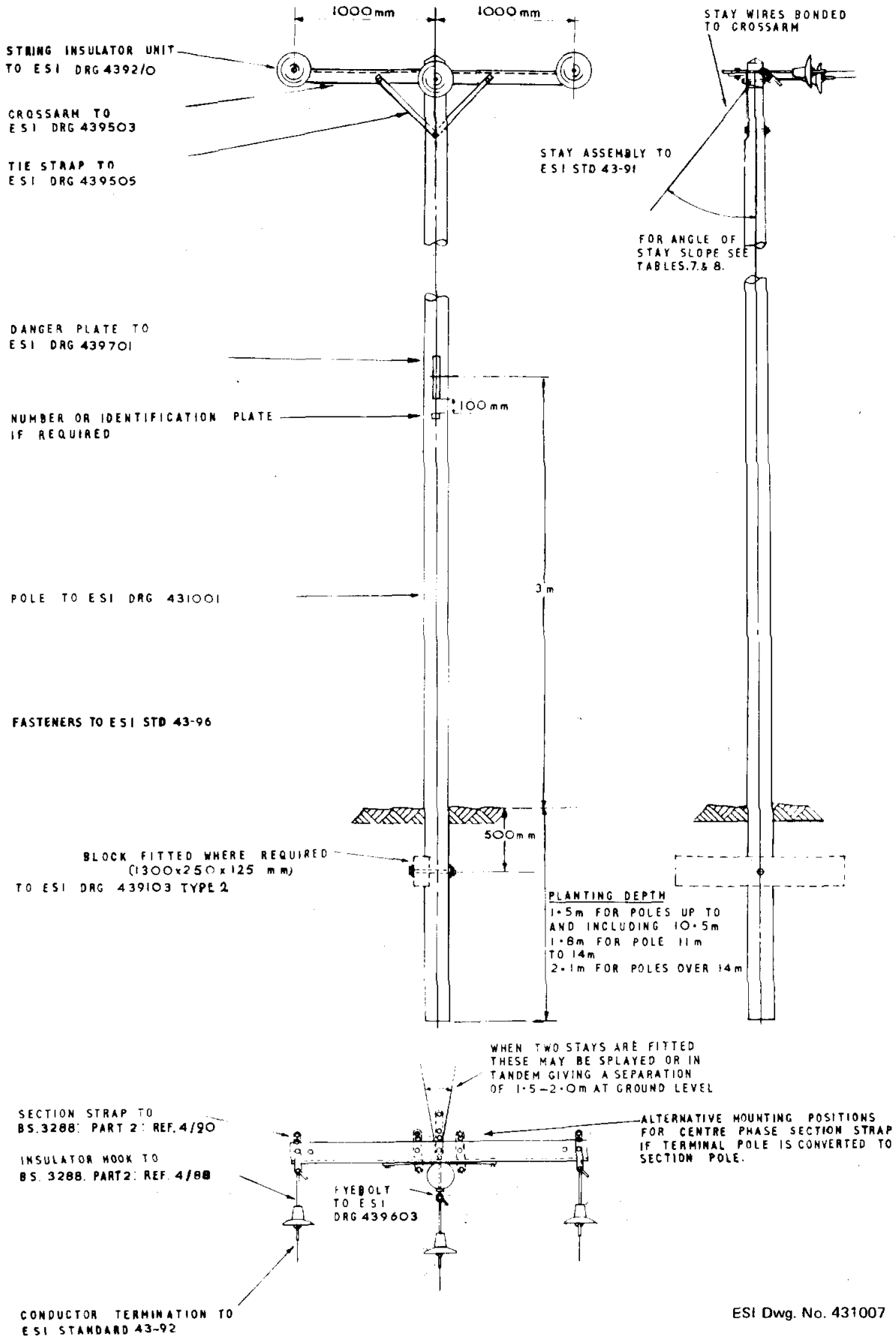
ESI Dwg. No. 431005

Fig. 5 General assembly section – angle pole up to 60°
(Limited to 56° for 50 mm² ACSR PVC: notes on Tables 7 and 8, page 20)



ESI Dwg. No. 431006

Fig. 6 General assembly section with gusset plate - angle pole between 60° and 75°
(Above 56° for 50mm² ACSR PVC: notes on Tables 7 and 8, page 20)



ESI Dwg. No. 431007

Fig. 7 General assembly – terminal pole

Fig. 8 General assembly - tee-off pole

APPENDIX A

LINES IN SEVERE ENVIRONMENTS

The purpose of this Appendix is to specify modifications to this Standard to cater for lines in situations where the design conditions specified in The Electricity (Overhead Lines) Regulations, 1970 may be exceeded due to geographical or weather conditions. A definition of such situations is not given because it is felt that this is best done by individual Electricity Boards based upon local knowledge and operating experience.

A1. SPAN LENGTHS

Recommended and maximum span lengths shall be in accordance with Table A1.

A2. CONDUCTORS

It is recommended that only 32 mm² copper or 50 mm² aluminium alloy or ACSR conductors be employed.

A3. SUPPORTS

A3.1 Section Angle and Terminal Supports

The strut loading imposed upon section angle and terminal poles shall be derived from Charts 1-12 as appropriate. These loadings shall then be related to the working loads of the poles by reference to Table A2, which is based upon a factor of safety of 3.5.

A3.2 Intermediate and Section Supports

When determining the diameters of intermediate and section supports, the wind loading span shall be that shown in Tables 2 and 3 multiplied by 0.80.

A4. STAYS

It is recommended that the largest angle of stay slope possible up to a maximum of 45° be adopted.

Table A1 - Recommended and Maximum Span Lengths - Lines in Severe Environments

	Recommended Span (m)	Maximum Span (m)
32 mm ² Copper	100	112
50 mm ² Aluminium Alloy and ACSR	100	112

Table A2 – Permissible Strut Loadings of Poles – Factor of Safety 3.5

Pole Planting Depths (m)	Pole Lengths (m)	Pole Top Dia. (mm)	Max. Strut Loading (kgf)		Pole Top Dia. (mm)	Max. Strut Loading (kgf)		Pole Top Dia. (mm)	Max. Strut Loading (kgf)	Dia. at 1.5 m (mm)	Pole Class (BS 1990)
1.5	8.5	125	1102		140	1413		150	1654	180	L
	9.0	125	956		140	1227		150	1438	180	
	9.5	125	880		140	1128		160	1536	185	
	10.0	125	777		140	997		160	1359	185	
	10.5	125	726		140	930		160	1264	190	
1.8	11.0	125	713		140	916		160	1250	195	
	11.5	125	671		145	931		165	1260	200	
	12.0	125	606		145	841		165	1139	200	
	13.0	130	596		150	817		170	1095	210	
1.5	8.5	150	2265		165	2791		180	3404	215	M
	9.0	150	2050		165	2524		180	3075	220	
	9.5	150	1872		165	2301		180	2801	225	
	10.0	150	1721		165	2114		185	2738	230	
	10.5	150	1594		170	2088		190	2689	235	
1.8	11.0	150	1550		170	2039		190	2635	240	
	11.5	150	1446		170	1898		190	2449	245	
	12.0	150	1356		170	1777		190	2288	250	
	13.0	160	1383		180	1790		200	2280	260	
	14.0	160	1292		180	1663		205	2231	275	
2.1	15.0	165	1337		185	1710		210	2281	290	
	16.0	170	1348		190	1709		215	2256	305	
1.5	8.5	190	5545		215	7306		240	9457	265	S
	9.0	190	5149		215	6765		240	8733	275	
	9.5	190	4661		215	6118		245	8289	280	
	10.0	190	4252		215	5575		245	7546	285	
	10.5	190	3905		220	5387		250	7255	290	
1.8	11.0	190	3773		220	5232		250	7079	295	
	11.5	190	3494		220	4837		250	6534	300	
	12.0	190	3252		220	4494		250	6062	305	
	13.0	195	3108		225	4252		255	5687	320	
	14.0	195	2849		225	3876		255	5158	335	
2.1	15.0	195	2716		225	3691		260	5138	350	
	16.0	200	2669		230	3594		265	4957	365	
Poles with top dia equal to min. specified in BS 1990.				Poles with mean value of top dia.				Poles with top dia. equal to max. specified in BS 1990.			

Notes

1. The above values of strut loading are safe working loads.
2. This table shall be used in conjunction with Charts 1-12.